Transportation Research Part E 159 (2022) 102630

Contents lists available at ScienceDirect

Transportation Research Part E

journal homepage: www.elsevier.com/locate/tre

Transport service selection and routing with carbon emissions and inventory costs consideration in the context of the Belt and Road Initiative

Yingxiu Qi [a], Steven Harrod [b], Harilaos N. Psaraftis [c], Maoxiang Lang [a,*]

[a] Key Laboratory of Transport Industry of Big Data Application Technologies for Comprehensive Transport, School of Traffic and Transportation, Beijing Jiaotong University, 100044 Beijing, PR China
[b] Department of Engineering Technology, Technical University of Denmark, 2750 Ballerup, Denmark
[c] Department of Technology, Management and Economics, Technical University of Denmark, 2800 Lyngby, Denmark

ARTICLE INFO

Keywords:
Belt and road initiative
Multimodal transportation
Green logistics
CR Express

ABSTRACT

The Belt and Road Initiative (BRI) entails investments to improve overland (rail) transport between Europe and China. This paper introduces a microscopic Multi-Commodity Flow Service Selection Problem for freight transport under the BRI and provides a decision tool for shippers to make door-to-door service plans. The minimizing objective function considers transportation costs, in-transit inventory costs, and carbon emissions. A series of sampled data of each provincial region of China are collected from Chinese multimodal transport operators. Results show that inland regions are strongly attracted to the rail mode for shipments to Europe. However, the "last mile" (including "first mile") transport from the shipper to the long-haul transport terminal strongly influences this choice, and carbon emissions are strongly influenced by the available last mile transport links. Under the dual impact of in-transit inventory and carbon emission costs, regions that prefer rail to maritime are much further east than suggested by previous literature.

1. Introduction

The *Belt and Road Initiative* (BRI) initiated by the Chinese government in 2013 is a political and economic cooperation strategy, which comprises the Silk Road Economic Belt and the Twenty-First Century Maritime Silk Road (OECD, 2020). Perhaps confusing for English speakers, railway lines are referred to as "belt" and deep-sea shipping lines that link the Far East to Europe are referred to as "road". To establish an ambitious Eurasian commerce and communication platform, significant capital investments along the "Belt" and "Road" have brought new and advanced transportation infrastructure with better service quality for international trade (Thürer et al. 2020, Ma et al., 2021).

Maritime logistics has always been the dominant transport mode in the Silk Road trade. About US$ 568 billion worth of products traveled via deep-sea shipping in 2017, 73.4% of the total export value from China to countries within the BRI. Similarly, about US$ 384.19 billion of import cargo (57.7%) was via maritime transportation (SIC, 2019). However, the recent COVID-19 pandemic has seriously disrupted the maritime container market: container freight rates in all world markets moved to historical highs between the

* Corresponding author at: School of Traffic and Transportation, Beijing Jiaotong University, No. 3 Shangyuancun Haidian District, Beijing 100044, PR China.
E-mail address: mxlang@bjtu.edu.cn (M. Lang).

https://doi.org/10.1016/j.tre.2022.102630
Received 7 April 2021; Received in revised form 8 January 2022; Accepted 27 January 2022
Available online 15 February 2022
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fall of 2020 and the spring of 2021 (UNCTAD, 2021) and the turmoil may last for years. The Shanghai Containerized Freight Index almost quadrupled in the period from 2020 Q2 to 2021 Q2 (DHL, 2021). In addition, the importance of rail corridors between China and Europe as alternatives to the traditional maritime routes has been rendered more significant by events such as the vessel blockage of the Suez Canal. In any event, recent developments have increased the need to consider alternative routes between the Far East and Europe as a means to keep transport costs under control and maintain transport resilience and redundancy. Even container lines such as MSC and Maersk have recently initiated rail services from the Far East to Europe and vice versa.

China Railway Express (CR Express), the railway services along the BRI, has experienced a steady demand growth in the last few years. The total value of cargo between Europe and China delivered by CR Express has increased from 1% in 2015 to 7% in 2020. By June 2021, over 40,000 CR Express departures connected Chinese cities and 168 destinations across 23 countries in Europe and Asia (NDRC, 2021). Most CR Express train journeys range from 9000 to 13,000 km. The average transit time from China to Southeast Asia is about 5 days, to central Asia about 10–18 days and to Europe about 12–23 days. The average transport time of CR Express from Chinese central cities to European seaport cities is one third that of maritime shipping (Jiang et al., 2018), and time saving has become one of the core advantages of CR Express.

In the context of BRI, this paper addresses the following questions:

1. What are the primary factors influencing the modal choice between rail and sea? What is CR Express's competitive advantage?
2. Many shipments have multiple feasible departure terminals. What are the factors that influence the choice of a terminal?
3. How do capacity and schedule limits affect mode and route? Should a shipment be held for the next available departure, or be routed to an alternate departure node?
4. Is there a regional pattern to the competitive position of transport modes and services?
5. There is increasing attention to carbon emissions and the pursuit of "green logistics". How will carbon pricing affect modal choice?

A clarification is in order. Considering that westbound traffic is the predominant trade direction of CR Express, this paper focuses on the traffic from China to Europe rather than the opposite direction. UNCTAD (2020) reports that westbound traffic has regularly been about double that of eastbound traffic in the period 2014 to 2020. This is mainly due to the physical trade imbalance between China and Europe (Wen et al., 2019). From the perspective of a shipper who ships from China to Europe, what happens on the return leg is not directly relevant, with the possible exception of the impact of backhaul cargoes on the general price level of service both ways. However, as these prices are exogenous inputs to our model, one can essentially consider the two problems (East to West and West to East) as decoupled. The West to East problem also involves empty container repositioning, which is an important problem, in and of itself, but whose consideration is outside the scope of the paper. Therefore, this paper focuses on the China-Europe direction, and the potential beneficiaries of this study could be Chinese exporters, European importers and international freight agents in BRI.

To address this problem, this paper formulates a version of the capacitated multi-commodity flow problem (CMCFP) which is referred to here as a *Capacitated Multi-Commodity Flow Service Selection Problem* (CMCFSSP). A mixed-integer program (MIP) is developed so as to assign shipments to the lowest cost (best utility function) service choice from a number of land and sea transport services.

The main contributions of this paper are as follows:

1. This paper presents a capacitated multigraph, which defines service arcs between two physical nodes by transport mode, departure date, transit time, cost, distance, and carbon emissions. It makes the CMCFSSP model microscopic with respect to choice of vehicle, train or ship.
2. The CMCFSSP model offers a more detailed MIP planning formulation which measures the objective from the date of shippers' service request instead of from shipment departure. It is further extended with simultaneous service network design with the service departure timetable as part of the decision variables in the formulation.
3. A numerical analysis of services preferences and their changes are offered under the dual impact of time value (in-transit inventory cost) and carbon emissions.
4. The analysis incorporates a treatment of CO2 emissions along the BRI, and it is seen that maritime shipping does not always emit less CO2 than inland railway transport. Shippers' behaviors are studied under different carbon rates.
5. This paper provides some suggestions under the BRI for shippers to route cargoes, for operators to improve the service network and create market segmentation, as well as for governments to determine carbon tariffs.

The rest of this paper is organized as follows: Section 2 outlines the related literature. Section 3 presents the logistics service process in the BRI and provides a novel definition of services. Section 4 introduces the Capacitated Multi-Commodity Flow Service Selection Problem (CMCFSSP). Section 5 presents a small-sized and a large-sized case study and performs a sensitivity analysis of key parameters of CMCFSSP. Section 6 presents a further study about departure time optimization, and Section7 offers a discussion of the results and the managerial suggestions for shippers, operators, and governments based on the prior sections. Finally, Section 8 summarizes the contents of this paper.

2. Literature review

The Belt and Road Initiative (BRI) is less than ten years old, so the related literature is relatively modest. The literature specific to CR Express is reviewed first, followed by a methodology review.

Table 1
Relevant quantitative research on the CR Express.

Research Topic	Literature	Description	Method	With time value	With carbon emission
Location	Zhao et al. (2018)	Consolidation center capacity evaluation and location	TOPSIS + MIP(U)		
	Ma et al. (2020)	Multimodal hub location with travel time restriction	MIP(U)	√	
	Zhang et al. (2020)	Nodes importance ranking	MADM		
	Sun et al. (2020)	Consolidation center location	TOPSIS		
	Tang et al. (2021)	Location routing decision of a hub-and-spoke network	BIP	√	
Subsidy	Kundu and Sheu (2019)	The effect of government subsidies on shippers' mode switching behavior	Game theory	√	
	Yin et al. (2020)	Freight subsidy mechanism	Bi-level programming	√	
	Ma et al. (2021)	Government subsidies and pricing strategies in a duopoly market	Game theory		
Choice behavior & Competitive	Li et al. (2020)	Customers' demand preference analysis and factors ranking	BWM + MCDA	√	√
	Wen et al. (2019)	Economic corridors selection under BRI	Utility theory	√	√
	Lu et al. (2019)	Spatial competition between rail and sea	Geometric distance calculation	√	
Others	Liu et al. (2018)	Supply chain coordination	Game theory		
	Jiang et al.(2018)	Hinterland Pattern of CR Express	Binary logit model	√	
	Yang et al. (2020)	The economic impact of CR Express	Utility theory	√	
	Wei and Lee (2021)	Coordinated alliance design between rail and sea	Entropy-weighted TOPSIS		
Service selection	**This paper**	**Service selection for shippers & service improvement for operators**	**MIP(M)**	√	√

Method: MIP(U): Mixed-integer programming for uni-commodity flow problem, MIP(M): Mixed-integer programming for CMCF, MADM: Multiple attributes decision making modal, BWM + MCDA: Best-worst method and multi-criteria decision analysis method, BIP: Binary integer programming

2.1. CR Express

Research on the CR Express is a very new research topic. Qualitative research on it was the mainstream several years ago, and it doesn't even have a recognized name. Before "CR Express" became the official title, keywords used in academic research included the TransAsia railway (Renliang, 2016), China-Europe Express (Wang, et al., 2017), China-Europe block train (Fu, et al., 2016) and Sino-Europe Regular Cargo Train (Peng and Ma, 2016), etc.

Quantitative research concerning CR Express emerged from 2018 as summarized in Table 1. Lu et al. (2019) formulate a macroscopic estimation based on distance, freight cost, and inventory cost that defines a boundary line between west and east China. On the west side of this line, shippers choose rail, and on the east side of this line, shippers choose maritime. Mou et al. (2021) analyze the spatial pattern of ports along the Maritime Silk Road and assert that the ports in the Straits of Malacca, the United Arab Emirates, the northern Mediterranean coastal region and the China-Japan region have advantageous locations. Wen et al. (2019) perform a macroscopic analysis of proposed investments in inland economic corridors of the BRI and identify four of the six corridors as a high priority for development. Zhao et al. (2018) apply centroid analysis to the question of where new CR Express terminal locations should be placed, while Zhou et al. (2021) use a gravity model to find new multimodal hubs locations from the view of the air transport network. As for service network improvement, Yang et al. (2017) use a bi-level programming model to optimize shipping lines between Asia and Europe, which considers the competition between the New Eurasia Land Bridge (one of the Silk Road corridors) and maritime shipping. Jiang et al. (2018) and Kundu and Sheu (2019) analyze the impact of the government subsidies on attracting more flows from maritime to CR Express, while an appropriate subsidy reduction scheme for CR Express is found to be more desirable for its sustainable development in the paper of Yin et al. (2020).

2.2. Methodology literature including carbon emission and inventory cost factors

The shippers' choices of transportation mode are typically based on the performances of services they desired (Crevier et al. 2012). In a multimodal transportation corridor with various service providers, cost and delivery time are the factors most often cited as determinants (Li et al., 2020). Andersen et al. (2009) present an international multimodal transportation problem that takes the transshipment time window as a core constraint to link the internal and external transportation system. Akgün et al. (2020) present a multimodal multi-commodity and multi-period linear programming model for Turkey's coal distribution system. Sun and Lang (2015) study a container shortest path problem within scheduled rail services and time-flexible road logistics services. Karsten et al. (2018) impose hard transit time restrictions on each container flow. Kazemzadeh et al. (2021) introduces a node-based lagrangian relaxation algorithm to solve the multi-commodity capacitated fixed-charge network design problem.

"Just in time" has placed a higher value on transport speed (Holguín-Veras et al, 2021), reducing flow time and inventory cost in

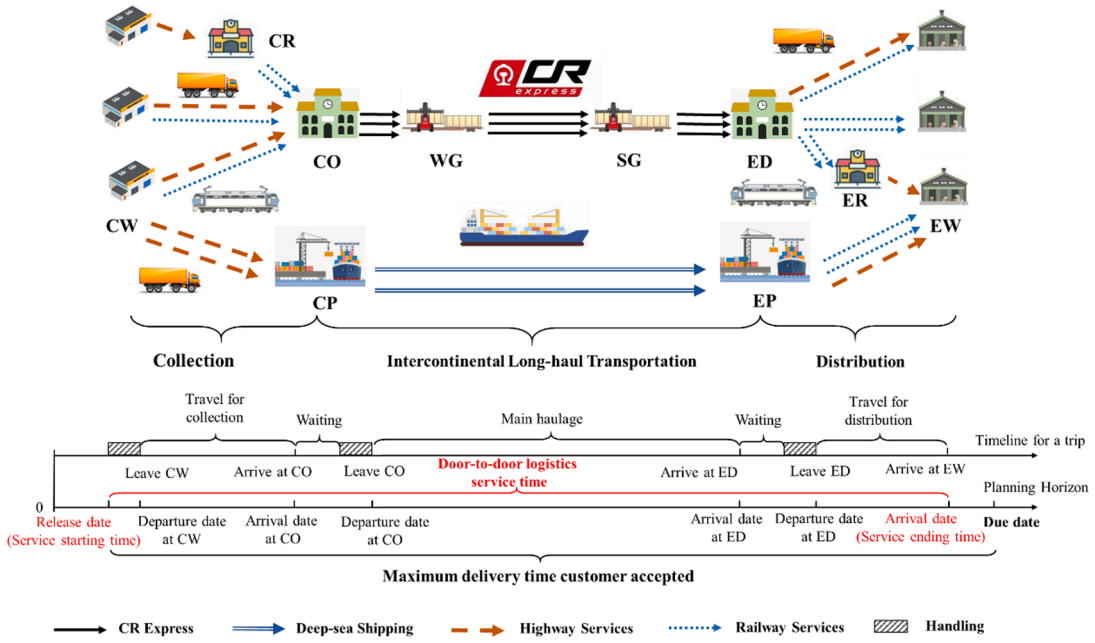


Fig. 1. The logistics service process in the Silk Road corridors.

manufacturing (Leuveano, et al., 2019). Time value or time restriction continues to rise in freight transportation (Small, 2012). Ma et al. (2020) measure the influence of transit time on the hub location of CR Express by a single commodity flow model. Psaraftis and Kontovas (2014) estimate the in-transit inventory cost in a liner shipping problem by multiplying delivery time by daily average Cost Insurance and Freight (CIF) value, presenting the capital influence of transportation cost, insurance, and cargo value. Tang and Li (2015) optimize routes by minimizing the average inventory cost and stock-out penalty in a China railway-based intermodal system.

Global warming introduces another pressing objective component: transport carbon emissions. Apple, a key client of CR Express, has been carbon neutral for its operations since April 2020, and one important consideration when Apple selects transport services is carbon emissions (Apple, 2021). Carbon emissions can vary significantly according to the type of vehicle, cargo loading, and traveling distance (Sun and Lang, 2015, Qu et al., 2016, Bauer et al., 2010). Psaraftis and Kontovas (2015) discuss the feasibility of slow steaming from various practical angles that could reduce fuel consumption directly. Jabali et al. (2012) use a non-linear formulation to account for carbon emission cost more accurately. Bektaş et al. (2019) summarize the operations research methods for green freight logistics including routing, scheduling, location and fleet deployment. It has become a consensus that maritime transport generates far less CO_2 than inland transport (Karsten, et al., 2015), however, travel distance is also a fundamental factor (Heinen and Mattioli, 2019). It is novel and necessary to study the geographical advantages for inland transport services under different carbon measures.

Looking at the above cited literature, and in the context of the BRI, we think that our paper is likely the first attempt to model shippers' service selections as a CMCFP in which arc costs include transport cost, in-transit inventory cost and carbon impact. It addresses a new transport service selection problem, CMCFSSP, by adding specific departures as decision variables. To measure the time value more accurately, the cost of transport is measured not from the time of departure, but from the time of release and request for transport. Last but not least, this paper is also the first attempt to develop an optimization approach that analyzes the environmental protection potential of inland rail versus maritime shipping. Shippers' behaviors are analyzed in detail under different carbon rates.

3. Service network statement

The CMCFSSP formulation is derived from a capacitated multigraph that includes nodes and directed arcs for origin collection, line haul, and destination delivery of containers. The single greatest complicating factor for the CR Express service is the need to change gauge from standard to wide gauge (WG) and then from wide to standard gauge (SG) at the east and west borders of Russia. This is the reason for additional node complexity in the network graph and the need for modeling dwell times.

Fig. 1 shows a service process where ports (CP, EP), rail stations (CR, CO, ED, ER), borders (WG, SG), and enterprises' warehouses (CW, EW) are connected by various services arcs. Flows that select land transportation begin from Chinese warehouses (CW), possibly via intermediary rail terminals (CR) to origin stations of CR Express (CO), and then transship to CR Express leaving China for Europe. and trains' destination in Europe (ED), may via railway station (ER), ending at European enterprises' warehouses (EW) by either road or rail transport.

Time line of a service process consists of the handling and waiting time at each node, as well as the traveling time between them. Note that, dwell times at container transfer nodes are variable. A faster delivery may leave for the subsequent node without waiting. Time is discretized in one-day increments. Service starting time (release date) at the origin warehouse in this paper is the moment

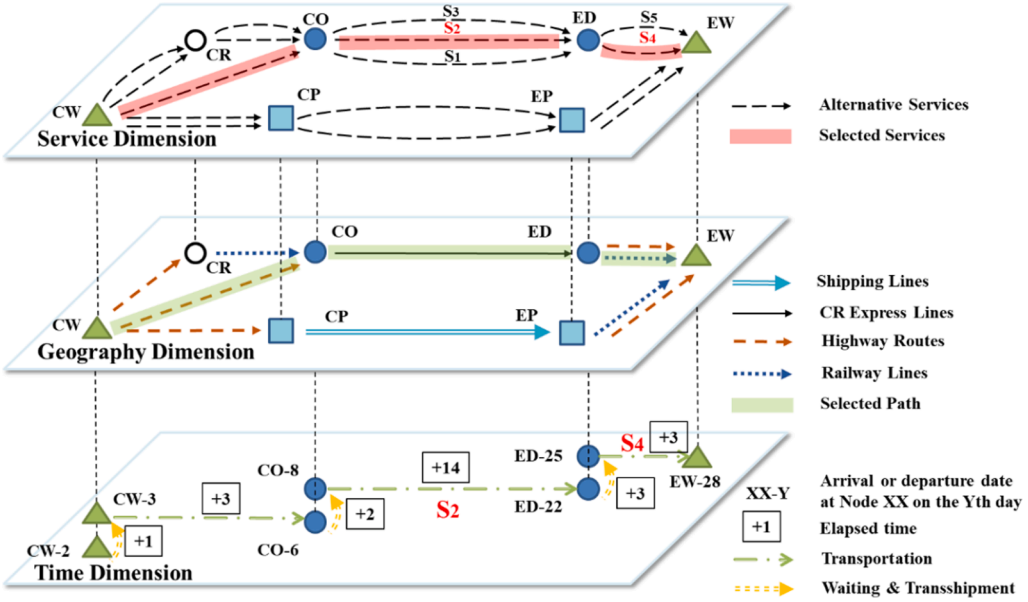


Fig. 2. A feasible door-to-door service plan of one shipment with two alternative corridors in the time–space-service network.

when cargoes are prepared to depart from the origin, not the departure date at origin as described in Sun and Lang (2015). The shipper's production plan and inventory strategy determine the release date and the due date.

The dashed orange arcs and dotted blue arcs in Fig. 1 correspond to roadway and railway collection and distribution services, while the solid black lines and double blue lines present the CR Express and maritime logistics services, respectively. Unlike previous papers that define services by transportation mode (Sun and Lang, 2015) and voyage (Balakrishnan and Karsten, 2017), we define services between two nodes by transport mode, departure date, traveling time, distance, price, service capacity and carbon emission. For instance, there are three CR Express lines between CO and ED, which depart on Monday (S_1), Wednesday (S_2), and Saturday (S_3) respectively. The benefits of this kind of definition include:

1. The formulation is microscopic in train or ship selection and not limited to the modal choice or the geographical routing. In practice, a shipper needs to know the specific train or vessel to get prepared for production and delivery.
2. Each service has individual characteristics such as departure date, service capacity and transport cost, which could depict customers' utility more accurately.
3. A CR Express train competes with not only multiple shipping lines but also with other CR Express trains, including those departing at the same station or other cities.

Lines in the geography dimension (middle layer) in Fig. 2 show a simplification of the actual transportation network, which depicts some physical routes for one OD pair (the pair CW-EW). Parallel dashed black lines in the service dimension (top layer) present groups of alternative logistics services. The pink highlights in the top layer present a complete service plan following CR Express train S_2 and railway distribution service S_4. The arrival date at ED is 22 days, calculated by adding the departure date 8 at CO to the transit time of S_2, 14 days. Likewise, the handling time window, total transit time limitation and capacity constraints cannot be ignored when scheduling service plans. Otherwise, shippers should change the departure date, transit time, transportation mode, or freight corridor by selecting other alternative services.

4. Capacitated multi-commodity flow service selection problem

The Capacitated Multi-Commodity Flow Service Selection Problem (CMCFSSP) is an extension of a CMCFP with a fixed service schedule. It determines the service sequence of each order, departure ports or railway stations, selected containerships or CR Express trains as well as arrival time at each node, etc. under the constraints of capacity and delivery time. Without loss of generality, the model assumes that service capacity is known. Each order has a single origin–destination pair and contains only one kind of commodity with demand, release date and due time known in advance. Table 2 summarizes the model notation.

4.1. *Objective function*

The objective function (1) minimizes the sum of direct transportation cost, carbon emission cost, and in-transit inventory cost. The first part is transportation cost for shipping cargoes by service s through arc (i,j); The second part is the carbon emission cost that equals

Table 2
Parameters and variables of CMCFSSP.

Indices	Description
i,j	Index of the nodes
s	Index of the transport services
k	Index of orders
(i,j)	Index of service arcs from node i to node j
Sets	Description
K	Set of orders, where $k \in K$
O	Set of origins presenting shippers' warehouses
o^k	Origin of order k presenting the shipper's warehouse, where $o^k \in O$
D	Set of destinations presenting receivers' warehouses
d^k	Destination of order k presenting the receiver's warehouse, where $d^k \in D$
N	Set of intermediate nodes, including ports, railway stations, etc. and $i,j \in N$
N^k	Set of intermediate nodes of order k, where $N^k \subseteqq N$
V	Set of nodes, where $V = O \cup N \cup D$
V^k	Set of nodes of order k, where $V^k = \{o^k\} \cup N^k \cup \{d^k\}$, and $V^k \subseteqq V$
A	Set of arcs from node i to node j, where $A = \{(i,j) \mid i,j \in V, i \neq j\}$
A^k	Set of arcs of order k, where $A^k = \{(i,j) \mid i,j \in V^k, i \neq j\}$, and $A^k \subseteqq A$
a^s	The arc where service s running, where $a^s \in A$
S	Set of services in service network connecting separate nodes
S_{ij}	Set of services departing from node i to node j where $(i,j) \in A, S_{ij} \subseteqq S$
Parameters	Description
b^s	Traveling distance of service $s \in S$ in km
h^s	Departure date (days) of service $s \in S$
l^k	Max delivery time of order $k \in K$ for the whole trip in days
p_i^k	Processing time of order $k \in K$ at node $i \in \{o^k\} \cup N^k$ in days
q^k	Size of order $k \in K$ in FEU
r^k	Shipper release date (days) of order $k \in K$
M	A big number
t^s	Traveling time of service $s \in S$ in days
u^s	Transport capacity of service $s \in S$ in FEU
u_i	Available handling capacity of node $i \in V$ in FEU
c^s	Transport cost of service $s \in S$ in US$/FEU
e	Carbon price in US$/$ton$
α^s	Carbon emission of service $s \in S$ in ton/(FEU·km)
β^k	The in-transit inventory cost of order $k \in K$ in US$/FEU per day
Decision variables	Description
x_{ij}^{ks}	A binary variable; 1, if order k uses service s through arc (i,j), and 0, otherwise
z_i^k	Non-negative variable; the arrival time of order k at node i

to the product of the carbon price (e) and carbon emissions, where carbon emissions are induced by delivery cargoes by service s via arc (i,j) for b^s kilometers; The last part depicts the in-transit inventory cost, the loss of income that the cargo owner bears while the cargo is in transit, which is adapted from Psaraftis and Kontovas (2014). Mathematically, the daily in-transit inventory cost is equal to $PR/365$, where P is the CIF value of the cargo and R is the cost of capital. In this paper, the daily unit in-transit inventory cost is expressed per FEU per day(β^k). As order k is released at the date r^k, the total in-transit inventory cost equals the daily cost (β^k) multiplied by the cargo volume (q^k) and the transit time of the order ($z_j^k - r^k$).

$$min \sum_{k \in K} \sum_{(i,j) \in A} \sum_{s \in S} c^s \cdot q^k \cdot x_{ij}^{ks} + e \cdot \sum_{k \in K} \sum_{(i,j) \in A} \sum_{s \in S} \alpha^s \cdot b^s \cdot q^k \cdot x_{ij}^{ks} + \sum_{k \in K} \sum_{j \in \{d^k\}} \beta^k \cdot q^k \cdot (z_j^k - r^k) \tag{1}$$

Both transportation and carbon emission costs are indexed to the selected services, influenced by transportation mode, travel distance, cargo volume, etc. The transportation costs are end-user prices, which include government subsidies, transshipment cost, holding cost at intermediate nodes and other miscellaneous expenses. The in-transit inventory cost is related to in-transit time, daily inventory cost and cargo volume, depicting the impact of travel time on service selection behaviors. It includes waiting costs at different stations or ports caused by mismatched service sequences.

It should be clarified that international freight transport is thus far exempt from the internalization of carbon emissions cost. However, with the development of decarbonized transport, there are more and more applications of the "Eurovignette" directive in EU road transport and more cities begin to charge congestion fees (for instance Singapore and London). Thus, considering carbon emission

cost is important to stimulate green logistics and to examine whether policies are moving towards low-carbon solutions.

4.2. Constraints for service network

4.2.1. Network flow constraints

Flow balance constraints ensure every order flows from its origin to its destination. Infeasible nodes of each order have also been removed in pre-processing, reducing the scale of this problem at the same time.

$$\sum_{h\in N^k\cup\left\{d^k\right\}|(j,h)\in A^k}\sum_{s\in S_{jh}}x_{jh}^{ks}=1,\forall k\in K,j\in\{\mathrm{o}^{\mathrm{k}}\} \tag{2}$$

$$\sum_{i\in\{o^k\}\cup N^k\left|(i,j)\in A^k\right.}\sum_{s\in S_{ij}}x_{ij}^{ks}=1,\forall k\in K,j\in\{\mathrm{d}^{\mathrm{k}}\} \tag{3}$$

$$\sum_{i\in\{o^k\}\cup N^k|(i,j)\in A^k}\sum_{s\in S_{ij}}x_{ij}^{ks}-\sum_{h\in N^k\cup\{d^k\}|(j,h)\in A^k}\sum_{s\in S_{jh}}x_{jh}^{ks}=0,\forall k\in K,j\in N^k \tag{4}$$

4.2.2. Service capacity

Node capacity is the maximum number of containers handled in a time unit (day), including loading capacity and transshipment capacity. u_i represents the available handling capacity at node i as input data.

$$\sum_{k\in K|i\in\{o^k\}\cup N^k}q^k\cdot\sum_{j\in N^k\cup\{d^k\}|(i,j)\in A^k}\sum_{s\in S_{ij}}x_{ij}^{ks}\leqslant u_i,\forall i\in O\cup N \tag{5}$$

Constraint (6) depicts the loading capacity of each service, determined by transportation mode and the size of truck, vessel, or trains deployed. Parameter u^s for maritime is 10,000 FEU, for highway 100 FEU, and for railway service 45 FEU.

$$\sum_{k\in K|(i,j)\in A^k\cap\{a^s\}}q^k\cdot x_{ij}^{ks}\leqslant u^s,\forall s\in S \tag{6}$$

Other constraints related to given schedules are implicit in data and show their impacts when shippers select transport services. For example, the transport capacity of one block determines the number of alternative services in this railway section. It is determined by the railway signaling system to avoid collisions and ensure efficient operation.

4.2.3. Transit time constraints

This model defines a door-to-door logistics service beginning from the moment when all the preparations for delivery are finished at shippers' warehouses, and then it records the arrival time at the origin warehouse as the service starting time. Note that there may be a waiting period between the service starting time and the departure time at origins.

$$z_j^k=r^k,\forall k\in K,j\in\left\{\mathrm{o}^{\mathrm{k}}\right\} \tag{7}$$

Constraint (7) sets the service starting time at origins, while constraint (8) presents the arrival time at intermediate nodes and destinations. It also determines the lower bound of arrival time, which is a function of the departure time and traveling time of the selected logistics service.

$$\sum_{i\in\{o^k\}\cup N^k|(i,j)\in A^k}\sum_{s\in S_{ij}}x_{ij}^{ks}\cdot(h^s+t^s)\leqslant z_j^k,\forall k\in K,j\in N^k\cup\{d^k\} \tag{8}$$

The arrival time of each order at its destination should be zero when there is no cargo flowing through these nodes. The arrival time at its destination should not exceed its due time plus the release time, either.

$$z_j^k\leqslant(l^k+r^k)\cdot\sum_{i\in\{o^k\}\cup N^k|(i,j)\in A^k}\sum_{s\in S_{ij}}x_{ij}^{ks},\forall k\in K,j\in\{d^k\} \tag{9}$$

Constraint (10) restraints the departure time and the departure service from nodes, which ensures that every order has to be prepared before the train's or ship's departure. Considering that each departure node connects with multiple arrival nodes and that each direction has several services available for one order, the arrival time at this node should not be zero directly when one direction fails to meet the hard time limitation. The second part on the right hand of the inequality makes it possible for this order to select another direction and its services.

$$z_i^k\leqslant\sum_{s\in S_{ij}}x_{ij}^{ks}\cdot(h^s-p_i^k)+M\cdot(1-\sum_{s\in S_{ij}}x_{ij}^{ks}),\forall k\in K,(i,j)\in A^k \tag{10}$$

M in constraint (10) is a large number, and coefficient reduction helps to avoid unnecessary computational pitfalls (Camm et al. 1990). Thus, M should be equal to l^k+r^k according to constraint (9).

Table 3
The maximum number of variables and constraints.

Variables or constraints	CMCFP model	CMCFSSP model
x_{ij}^{ks}	$K \times A$	$K \times S$
z_i^k	$K \times N$	$K \times N$
Service capacity constraints	A	S
Departure time constraints	$K \times A$	$K \times S$

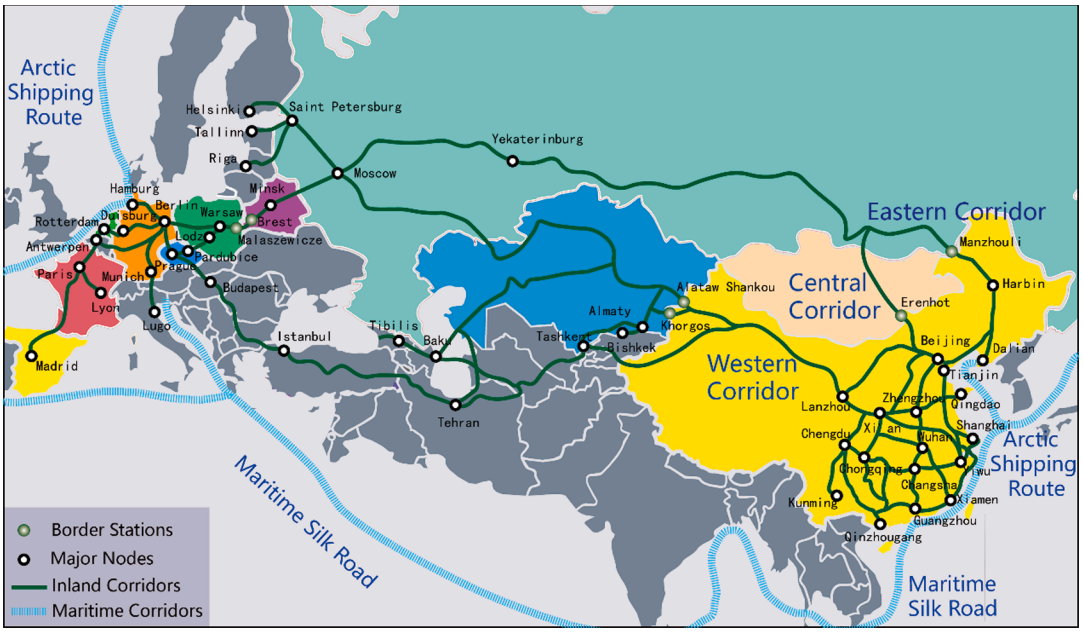


Fig. 3. The multi-modal transport network of the Silk Road corridors.

4.2.4. *Variable domains*

$$x_{ij}^{ks} \in \{0,1\}, \forall k \in K, (i,j) \in A, s \in S_{ij} \tag{11}$$

$$z_i^k \geqslant 0, \forall k \in K, i \in V^k \tag{12}$$

4.3. *Differences between the CMCFP and the CMCFSSP model*

The new definitions of release date and service make this CMCFSSP model different from the CMCFP model. First of all, to describe the time utility more accurately, the release date in this paper doesn't equal the departure time at origin, and thus the maximum arrival time in constraint (9) equals the release date plus the due time. Secondly, as each node is connected with multiple directions and each direction has multiple services arcs, a new parameter a^s is introduced in constraint (6) to depict which orders could select this service (service capacity). So is the big M part of constraint (10).

The new definition of services can make shippers' service decisions clear to a specific shift, but adding an additional dimension of services increases the complexity of this problem. Complexities of the CMCFSSP and the classical CMCFP are determined by four parameters, namely, the number of orders K, nodes N, arcs A, and service S, respectively.

Table 3 shows the maximum scales of the above two models. The corresponding number of services for each physical arc is critical for complexity. One arc in the CMCFP model represents one service (A = S), while more alternative services are added to each arc in the CMCFSSP model (A $\leq$ S). To reduce the scale of this CMCFSSP model, candidate nodes and arcs for each order (N^k and A^k) are introduced. Infeasible nodes and circuitous directions of each order have been removed in pre-processing.

5. An analysis of the Silk Road corridors

There are five transportation corridors under BRI connecting the Far East, Central Asia, and Europe, named Silk Road corridors. These corridors link the Trans-European Transport Network with China, consisting of three inland corridors and two maritime corridors as shown in Fig. 3.

Table 4
Sample case data.

Order	Origin	Destination	Due date (days)	Volume (FEU)
1	Beijing	Budapest	26	34
2	Yinchuan	Hamburg	30	26
3	Shanghai	Copenhagen	50	18
4	Zhengzhou	Athens	35	42
5	Wuhan	Prague	30	32
6	Guiyang	Madrid	40	24

Table 5
Feasible routes and nodes of each order via the Silk Road corridors.

Order	Maritime routing	Rail routing
1	Beijing- Tianjin-Hamburg-Budapest	Beijing-Jinan-Budapest
	Beijing- Tianjin- Piraeus -Budapest	Beijing-Xi'an-Budapest
		Beijing-Shenyang-Hamburg-Budapest
2	Yinchuan-Qingdao-Hamburg	Yinchuan-Lanzhou-Hamburg
	Yinchuan-Tianjin-Hamburg	Yinchuan-Xi'an-Hamburg
	Yinchuan-Shanghai-Hamburg	Yinchuan-Urumchi-Hamburg
3	Shanghai-Hamburg-Copenhagen	Shanghai-Suzhou-Duisburg-Copenhagen
	Shanghai-Aarhus-Copenhagen	Shanghai-Yiwu-Minsk-Copenhagen
		Shanghai-Hefei-Hamburg-Copenhagen
4	Zhengzhou-Lianyungang-Piraeus-Athens	Zhengzhou-Munich-Athens
	Zhengzhou-Shanghai-Piraeus-Athens	Zhengzhou-Urumchi-Lodz-Athens
	Zhengzhou-Ningbo-Piraeus-Athens	
5	Wuhan-Shenzhen-Koper-Prague	Wuhan-Pardubice-Prague
	Wuhan-Shanghai-Koper-Prague	Wuhan-Urumchi-Nuremberg-Prague
	Wuhan-Ningbo-Koper-Prague	Wuhan-Lodz-Prague
6	Guiyang-Shenzhen-Algeciras-Madrid	Guiyang-Urumchi-Madrid
	Guiyang-Guangzhou-Algeciras-Madrid	Guiyang-Chengdu-Tilburg-Madrid
	Guiyang-Fangchenggang-Algeciras-Madrid	

CR Express trains, such as the service from Shenyang to Hamburg, depart from the northeast of China via the *Eastern Corridor* through Manzhouli or Suifenhe border station connecting the Russian Siberian railway lines with European countries. The *Central Corridor* of CR Express links China via Erenhot pass to Mongolia, Russia, Belarus, and Europe, undertaking part of traffic flows to Russia and East Europe, including routes Zhengzhou-Hamburg and Chongqing-Duisburg). The *Western Corridor* is one of the busiest corridors that delivers cargo between Europe and China via Alataw Shankou and Khorgos passing through Kazakhstan, Russia, Belarus, Poland into other European countries, including routes Chengdu-Lodz, Yiwu-Madrid, and Wuhan-Pardubice.

The *21st Century Maritime Silk Road* refers to two main maritime corridors. The primary one connects China with Europe via Southeast Asia, the Indian subcontinent, and the Middle East. Shipping lines pass through the Malacca Strait and the Red Sea to the Mediterranean Sea. Ships can berth at ports in Greece, Italy, France, and Spain, where cargo often is taken further inland by road or rail, as far as the United Kingdom or Scandinavia. The other corridor is the *Arctic Shipping Route*, which shortens voyages significantly but transport capacity is limited due to thick ice and challenging straits during much of the year. Thus, this maritime route is not considered further in this study.

Freight forwarders play a significant role as users in the international transport process. They organize logistics services according to commercial contacts, including door-to-door, door-to-station, station-to-station, station-to-door, door-to-port, port-to-port, or port-to-door. Railway freight forwarders or logistics subsidiaries of railway companies take responsibilities for collecting cargo, booking CR Express services, and delivering them to customers, including DB SCHENKER, KTZ Express, TransContainer, and CR Express liner companies. Shipping companies, like A.P. Møller – Mærsk and COSCO Shipping Lines Company, operate a handful of shipping lines to enrich the maritime transport market. At the same time, some maritime shipping companies, like COSCO, also cooperate with railway companies and book inland services on behalf of their customers. The service decision tool provided by this paper could benefit these above freight forwards.

In this paper, models have been solved with the CPLEX version 12.9 by using a laptop with Intel Core I5 1.60 GHz processor and 8 GB memory. Termination condition was set to find the optimal solution of each problem.

5.1. Validation with a small data set

The small data set consists of six orders originating from Chinese cities and terminating in European cities. Cargoes flow in the 4 railway or maritime corridors from east to west as shown in Fig. 3. The experimental service network consists of 37 nodes, 92 physical arcs, and 133 alternative services. There are 320 constraints and 55,764 variables in total. The transport demands of these orders are shown in Table 4, and alternative routes for each order are given in Table 5.

The distances and transit time presented in **Appendix A** were collected from published schedules and direct inquiries. The freight

Table 6
Solution of small data set of 6 orders.

Cost type	Cost (US$)	Percent (%)
transportation	678,494	80.39%
inventory	130,121	15.42%
carbon emission	35,397	4.19%
total	844,013	100.0%

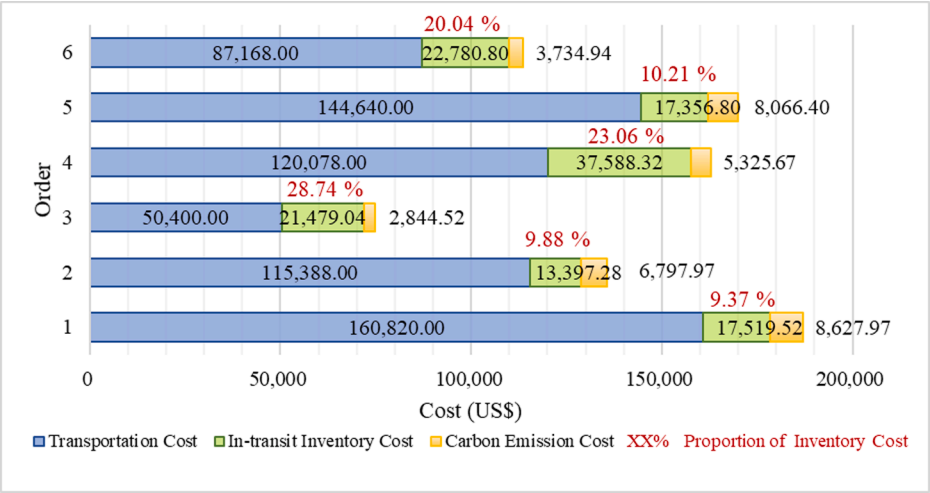


Fig. 4. Cost breakdown analysis of each order in **small data set.**

Table 7
Detailed routing from solution of small data set of 6 orders.

Order	Due time	Transit time	Selected route	Mode	Arrival dates at subsequent nodes
1	26	19	Beijing - Jinan	Road	3
			Jinan - Budapest	Rail	19
2	30	19	Yinchuan -Xi'an	Road	3
			Xi'an - Hamburg	Rail	19
3	50	44	Shanghai - Copenhagen	Ship	44
4	35	33	Zhengzhou - Lianyungang	Road	2
			Lianyungang - Piraeus	Ship	31
			Piraeus - Athens	Rail	33
5	30	20	Wuhan - Pardubice	Rail	17
			Pardubice - Prague	Rail	20
6	40	35	Guiyang - Fangchenggang	Rail	3
			Fangchenggang - Algeciras	Ship	31
			Algeciras - Madrid	Rail	35

collection information of the Chinese railway services comes from the published operation plan (CSRG, 2020). The information of vessel schedules is collected from shipping companies, like COSCO (COSCO, 2020), and the operation plan of CR Express trains is obtained from China Railway Container Transport Corporation. Motorway travel time and distances are obtained from public sources such as Bing. Typically, customs inspection, documents delivery time, and other transactions are included in the publicly quoted service travel time, so for lack of better information, the loading and unloading time at every transshipment node is 1 day. The release date of each order is assumed to be zero.

The current retail prices for railway services can be found on the China State Railway Group Co. Ltd.'s freight service website[4]. These are the end-user prices, which include government subsidies and miscellaneous expenses. The motorway prices are based on 13.5-meter-long truck-load service from the official website of the China Highway Logistics Freight Index (CFLP, 2021). These prices are location-specific and influenced by their local market. The loading capacity of a 13.5-meter-long truck is similar to a 40-foot container. The income loss in the base model has been set to US$ 450/ton per year as described by Psaraftis and Kontovas (2014), so β^k is US$ 27.12/FEU per day. The baseline carbon cost is assumed to be equal to the average European Emission Allowances price of April 2020, US$ 27.35 per ton (EEAP, 2020). It should be noted here that the above price, being market-driven, is subject to fluctuation. For that reason, a sensitivity analysis on this price is also conducted.

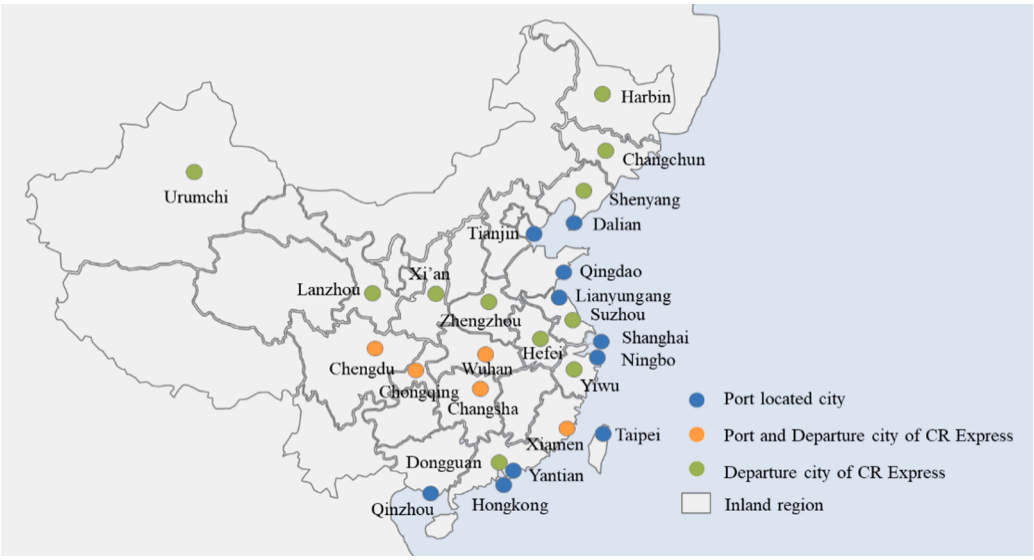


Fig. 5. Departure cities for main haulage transport in large-sized test.

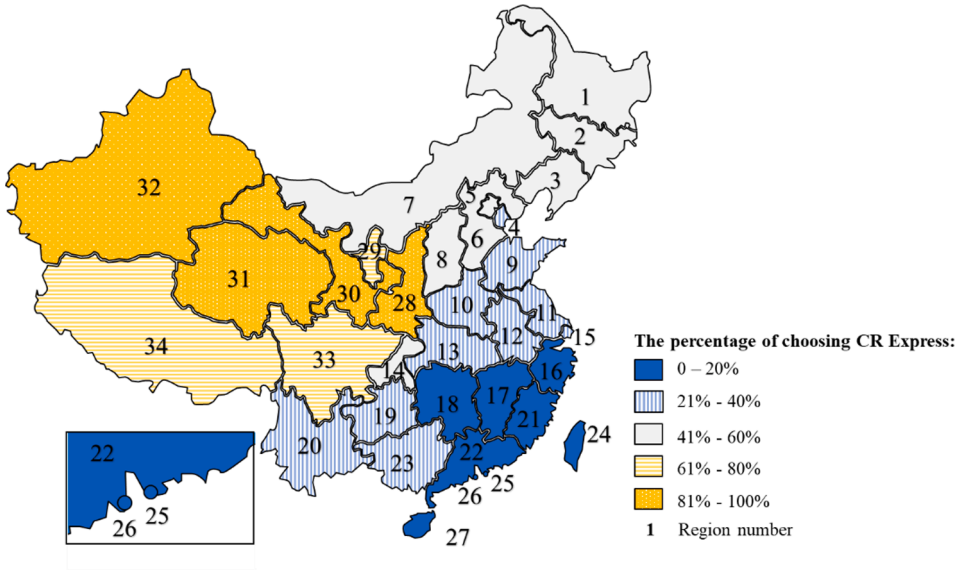


Fig. 6. Detailed mode selection results for 100 replications.

Psaraftis and Panagakos (2012) provide the average carbon emission benchmark for Silk Road corridors, 41.0 g/tkm by railway and 12.5 g/tkm by deep-sea shipping. Psaraftis (2016) presents sample European corridor emissions of 68.8 g/tkm and 46.51–71.86 g/tkm, and this model selects 68.8 g/tkm as the baseline value. For the unit of measure conversions, a 40-foot container (FEU) is assumed to be 22 metric tons.

Results for the solution of the small data set are shown in Table 6. Transportation is the dominant cost in the supply chain (excluding acquisition cost), followed by the in-transit inventory cost and carbon emissions.

Fig. 4 shows a detailed cost breakdown by order. As for orders 3, 4 and 6, which select maritime shipping, the in-transit inventory cost holds a greater proportion than other orders, more than 20%. In this small sample, inventory cost is higher for maritime by a ratio of 2:1.

The detailed routing for the optimal solution is shown in Table 7, including the selected routes and transit time information of each order. Orders 1, 2, and 5 are assigned to CR Express while the rest are routed by maritime, in this case an even division between rail and ocean transport. The quantity split is also even, with 92 FEU traveling by rail and 84 FEU traveling by sea.

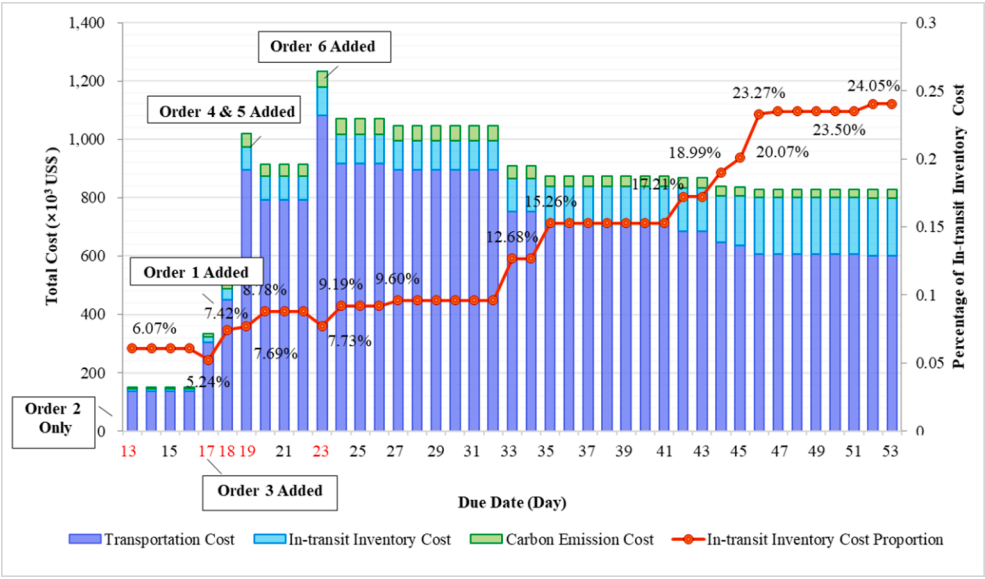


Fig. 7. Optimal cost structure with different due dates.

5.2. *Larger, realistic data set*

A random sample of size 100 is selected from 34 OD pairs. The resulting service network consists of 82 nodes, 123 physical arcs, and 457 alternative services linking 34 Chinese provincial regions with Hamburg. It takes about 1.5 h for 100 replications and the average time for each replication is about 54 s. Each replication has 1081 constraints and 1,618,603 variables.

Fig. 5 shows the departure cities of CR Express and maritime in the large-sized test. The origin cities in this test are selected from coastal and inland areas covering each provincial region, making it possible to analyze the service preferences in different areas.

An upper limit per order is set at 45 FEU because the capacity of an individual CR Express train is 45 FEU. Order size is simulated from a uniform distribution of U(1, 45), rounded to integers. The underway time from the Far East to Europe is usually 34–45 days by maritime and 12–25 days by train (OSJD, 2018). Taken together, these transport segments sum to an upper limit of 30–60 days for shipment release to destination delivery. Due dates are drawn from a uniform distribution of U(30, 60). The in-transit inventory cost of each order is sampled uniformly from U(10, 50). Finally, the free node capacity is sampled uniformly from zero to their theoretical capacity.

Different colors in Fig. 6 show the different proportions of orders that select CR Express in total, reflecting the degree of geographic preference of CR Express versus maritime shipping. Dark yellow presents the strongest preference of CR Express, while dark blue marks the advantage area for deep-sea maritime services. There is no obvious preference for areas in gray.

Fig. 6 shows that for this data set proximity to the ports including Tianjin (in region 4, short for 4), Qingdao (9), Shanghai (15), Ningbo (16) and Yantian (22), etc. is a major factor in the modal choice. The further away from these ports, the more competitive rail service is. Geographical advantages make Xinjiang (32) and other regions in dark yellow prefer CR Express to surrounding areas, which makes the inland routes to Hamburg many days shorter than the average route. The Northeast of China shows no preference to neither CR Express nor maritime shipping. Although regions (1), (2) and (3) are also close to the Pacific Ocean, the travel distances of CR Express trains that depart from these areas are less than 10,000 km, shorter than those in other east coastal areas. Additionally, good operation performance makes individual exceptions. CR Express trains depart from the region (28) with high service frequency and competitive price make the percentage of choosing CR Express bigger than the region (29).

5.3. *Sensitivity analysis*

Orders in sections 5.1 and 5.2 have individual transport demands, different in-transit inventory costs and different due times and the results depict the general situation at the whole country level. To study the selection preference under different scenarios and provide more rational managerial suggestions, the rest of this section (and specifically sections 5.3.1, 5.3.2 and 5.3.3) performs a sensitivity analysis on three basic parameters: transit time limitation, in-transit inventory cost and carbon emissions.

5.3.1. *Transit time limitation (Due dates)*

This section shows the impact of changes in due dates on service selection. Fig. 7 presents a detailed set of results for six orders of the small data set, with l^k incremented in days and discusses the changes of the total cost when keeping the other parameters fixed. Bars in Fig. 7 mean the total cost of transport cost, in-transit inventory cost and carbon emission cost, while the red line presents the percentage of in-transit inventory cost as a fraction of the total cost.

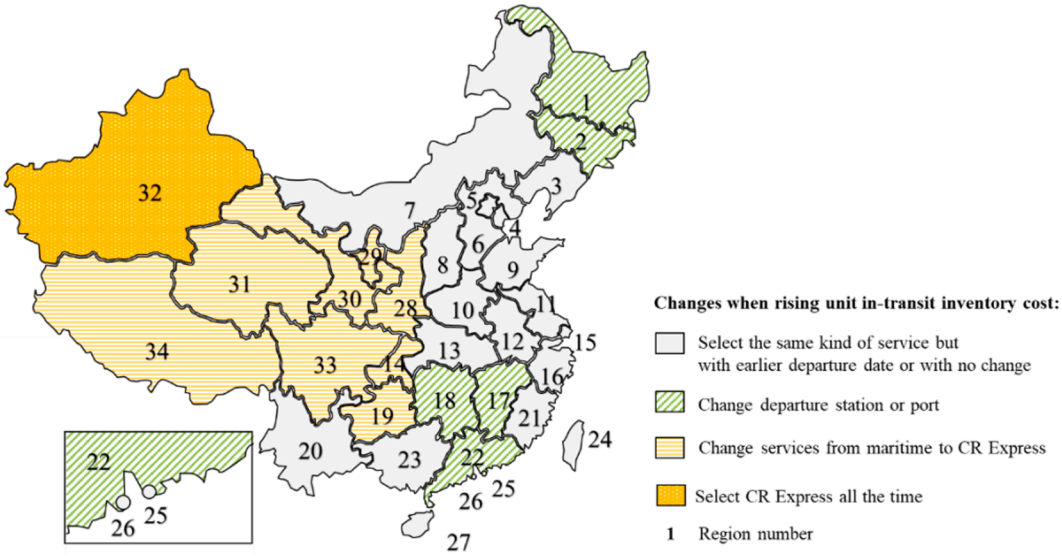


Fig. 8. Change in service selection, inventory cost changes from US$ 0 to US$ 27.12 /FEU per day for 100 replications.

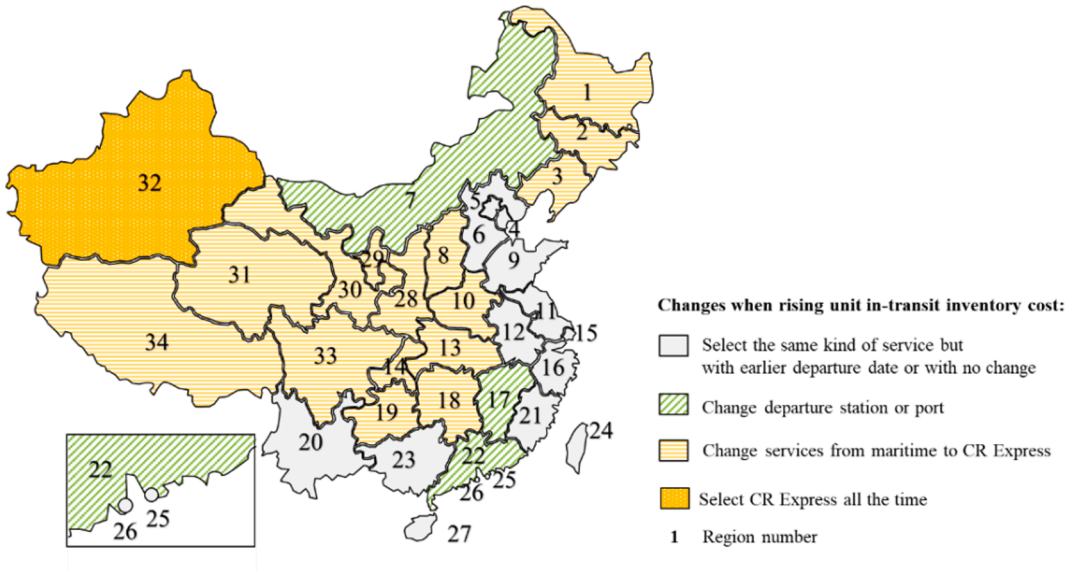


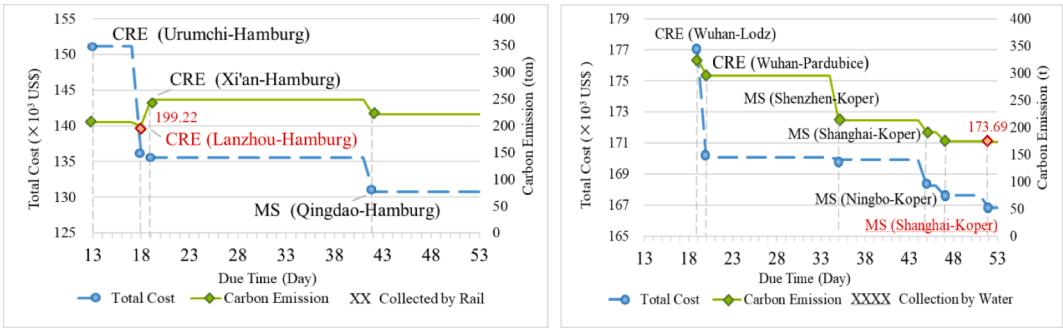
Fig. 9. Change in service selection, inventory cost parameter changes from US$ 0 to US$ 45.21/FEU per day for 100 replications.

As the delivery deadline is relaxed, it is seen that slower but cheaper logistics services are selected. An increase in total cost means that more orders are added into the feasible solution pool, while the decrease presents the change of service selection. For instance, a feasible solution of order 1 appears until the due date equals to 18 days. The decrease after 23 days shows that a cheaper service sequence is obtained, and the new service sequence may include a cheaper main haulage service or change to a cheaper collection or distribution service in general. What is to be noticed is that one-day relaxation at turning points between two service plans may entail a large cost saving.

It should be noted that there is always a negative correlation between transportation cost and in-transit inventory cost. High-value cargo (for instance, IT products) with higher in-transit inventory costs usually needs faster delivery. Cheaper transportation services sometimes mean slower delivery, but a longer transit time can result in inventory management pressure and worse market responsiveness. Thus, the inventory cost proportion of total cost increases from 6.07% to 24.05% when due dates are relaxed.

5.3.2. In-transit inventory cost

The sensitivity to in-transit inventory cost is evaluated geographically by solving the large data set at three levels of inventory cost. These levels are chosen from Psaraftis and Kontovas (2014), where the cargo values are US$ 0, US$ 15,000, and US$ 25,000/ton,



(a) Total cost and carbon emission for order 2 (b) Total cost and carbon emission for order 5

Fig. 10. Total cost and carbon emission of optimal solutions in the small-sized test with a range of due time.

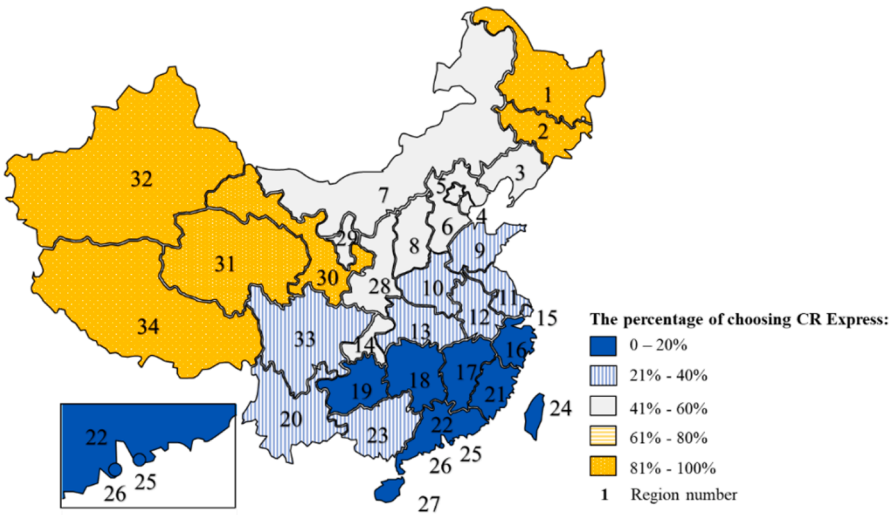


Fig. 11. Service selection for the large data set under a carbon emission minimizing objective.

corresponding US$ 0, US$ 27.12, and US$ 45.21/FEU per day, respectively. A zero value represents the case in which in-transit inventory costs are not considered. Three large-sized tests with controlled variables are made to show the service selection changes when the in-transit inventory cost is increased. Different scenarios are presented in Fig. 8 and Fig. 9.

From these results, it can be seen that cargo value (in-transit inventory cost) alone is not enough to motivate all traffic to select rail, but it does encourage orders to choose faster delivery. The higher the in-transit inventory cost (the higher the value of cargoes), the more popular the CR Express is. Departure nodes with matched services schedules (shorter waiting time) and shorter main-haul routes are the first choices. Specifically, shorter main-haul routes are equivalent to replacing a part of the journey of rail (CR Express) or water (maritime shipping) with road to some extent. For instance, orders in Hunan (region 18) give up departing from Shanghai port in region 15 and select shipping lines in Yantian port in region 22 when the unit cost rises to US$ 27.12 /FEU per day. Therefore, shippers would like to deliver cargo to southeast ports or northwest railway stations via faster services to reduce total transit time.

The coastal regions in the Northeast of China are notable for shifting to rail. Although these areas are close to the Pacific Ocean, Dalian port (region 3) is more than 2,000 km further away from Hamburg than Yantian Port (region 22). At a top ship speed of 25 knots, this is at least 44 h further away and longer if the ship makes additional port stops en route. On the other hand, these regions are about 1,000 km closer to Europe by rail than cities in southern China.

5.3.3. Carbon emissions

Generally, maritime services can offer very low carbon emissions per ton·km, particularly when slow steaming. However, travel distance is also a significant factor affecting carbon emissions (Heinen and Mattioli, 2019). Considering the travel distances by rail are much shorter than those by sea, the rail journey is still competitive with respect to carbon emissions in many instances. Analysis in this section is divided into three parts.

The first part of analysis focuses on finding the lowest-carbon service plans for orders with relaxed due time. Considering that tight due time will enforce orders to select CR Express, the due time of six orders in the small case are relaxed here to allow free competition

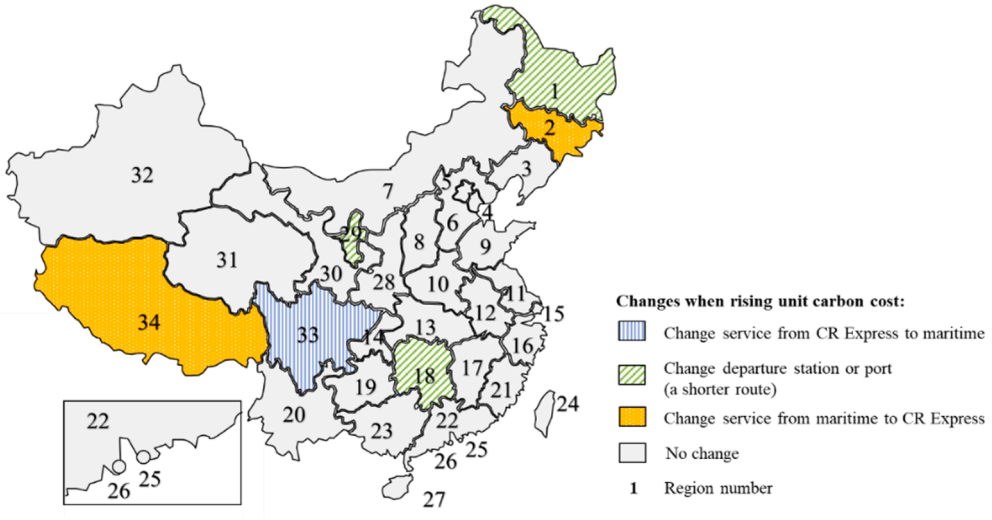


Fig. 12. Service changes for the large data set when increasing carbon cost from US$ 0 to US$27.35/ton.

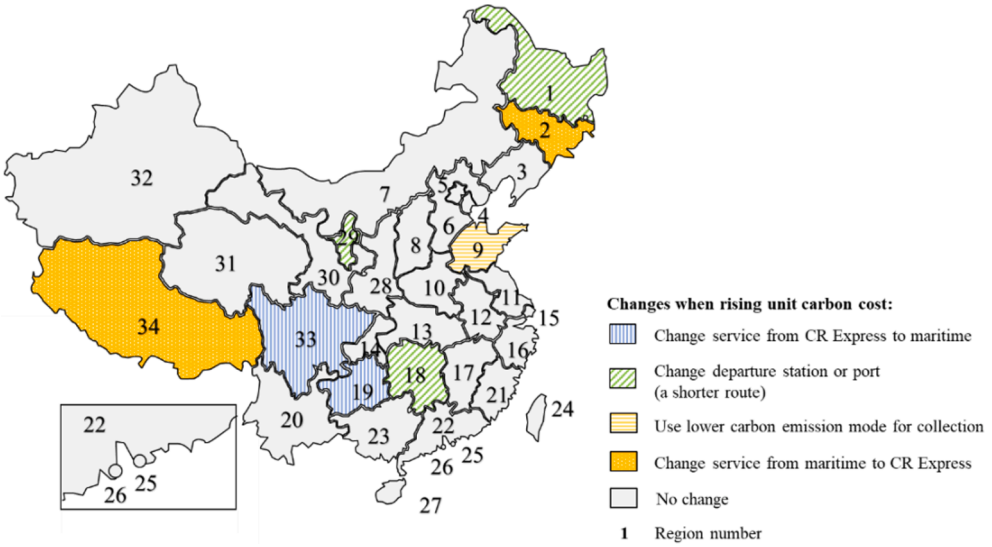


Fig. 13. Service changes for the large data set when increasing carbon cost from US$ 0 to US$ 100/ton.

between rail and water. The total cost and carbon emission of each optimal solution are recorded in Fig. 10 that shows the carbon emission fluctuations when selecting different services.

The carbon emissions of CR Express are not always higher than maritime. The CR Express (Lanzhou) in order 2 performs best for lower-carbon transportation, as shown in Fig. 10 **(a)**, generating about 199.22 tons of CO_2 for a door-to-door service plan. The reason why the order 2 shows its exception is that its origin city Yinchuan (29) is the only city in the northwest of China in the small-sized test. It is far from eastern ports. Roadway or railway services are needed to deliver cargo from origin to departure ports. What's more, the maritime voyage is more than twice of CR Express mileage. The geographical advantages of CR Express trains are also highlighted here. Unfortunately, the carbon rate is not high enough to prevent long-distance collection transport, and maritime shipping is still the best choice for shippers when the due time is larger than 41 days.

Fig. 10 **(b)** shows the impact of due time on the total cost and carbon emission of order 5. A tiny relaxation of due time allows shippers to select slower but cheaper services with lower carbon emissions. Shippers prefer delivering cargo to Shanghai by rail and loading on the container vessel (Shanghai-Koper), while they could select to deliver cargo to Shanghai via the Yangtze River if there is a one-week buffer. The sensitivity analysis of other orders is presented in **Appendix B**.

Secondly, do all regions in the Northwest of China prefer CR Express when it comes to green logistics? A more general geographical focus on carbon emissions is obtained by modifying the objective function to only consider carbon emissions, and deleting transport cost and inventory cost.

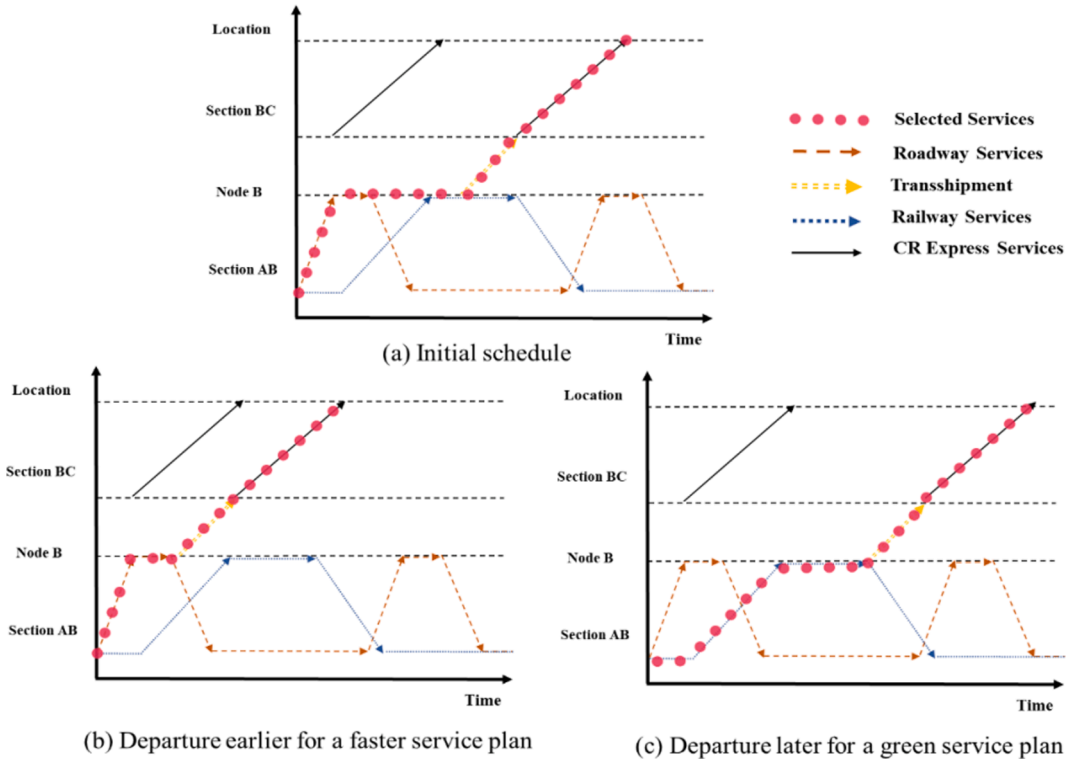


Fig. 14. The impact of departure date on service selection.

Fig. 11 presents the results for the large data set (with a baseline carbon cost of US$ 27.35 per ton). Regions 1, 2, 30, 31, 32, and 34 become dominated by rail services and the short distance is the main reason for preference. Note that, even though maritime shipping is obviously more environment-friendly than CR Express in the southeast of China, the percentage of choosing maritime shipping is not 100%. This is because only CR Express could meet the tight transit time limitation. So is CR Express in the northwest of China. The percentage of choosing rail or sea of one region depends on the due time and the transit time of its alternative services.

At the end of 2019, there were about 133,000 electric locomotives in China, approximately 63.4% of the locomotive fleet. The electrified rail network is more than 100,000 km, decreasing the comprehensive energy consumption per unit of transportation workload by 12.6% since 2014 (NRAC, 2019). It would be expected that advances in carbon reduction for the rail journey would push the decision boundary closer to the coast.

Finally, to reflect market fluctuations of the carbon price and evaluate sensitivity to various political proposals for a carbon pricing scheme, this model is further solved at three levels of carbon price: US$ 0, US$ 27.35, and US$ 100 per ton of carbon emissions. This range of values is motivated by the "High-Level Commission on Carbon Prices and Competitiveness" report (CPLC, 2019). This report estimates that the carbon price should be US$ 50-US$ 100 per ton by 2030 to drive a substantial change. It is also noted that the Marshall Islands and the Solomon Islands have recently recommended to the IMO a carbon price of US$100 per ton of CO_2 in the context of the discussion on market-based measures (Marshall, 2021).

Modal shift changes when carbon costs change are presented in Fig. 12 and Fig. 13. Note that there are more options than just shifting from maritime to rail in western areas or shifting from rail to maritime in eastern regions. There are also incremental changes such as shifting the local transport from road to rail, and choosing more direct routes (which may be more expensive). Additionally, highway services are more sensitive to carbon emission costs than railway services. The higher carbon price encourages a change in local ("last mile") transport by shifting cargo from road to rail or inland waterway, or decreasing driving distances by selecting a closer departure port (CP) or station (CO). For instance, orders in Hunan (18) shift to Yantian port (22) rather than Shanghai port (15) to transship to vessels, when increasing the unit carbon price from US$ 0 to US$ 27.35. Therefore, the driving distance decreases from 1031 to 784 km. When carbon price increases from zero to US$100, Shandong (9) selects railway instead of highway services to deliver cargo to Qingdao port (9), even though it requires more transport time. All numerical results for all the above scenarios are presented in the **Appendix C**.

6. Further extensions

In the previous section, the service selection method is presented from the perspective of shippers, and the competition between the CR Express and maritime shipping in the Silk Road corridors is quantified from a range of parameters. In this section, the model is

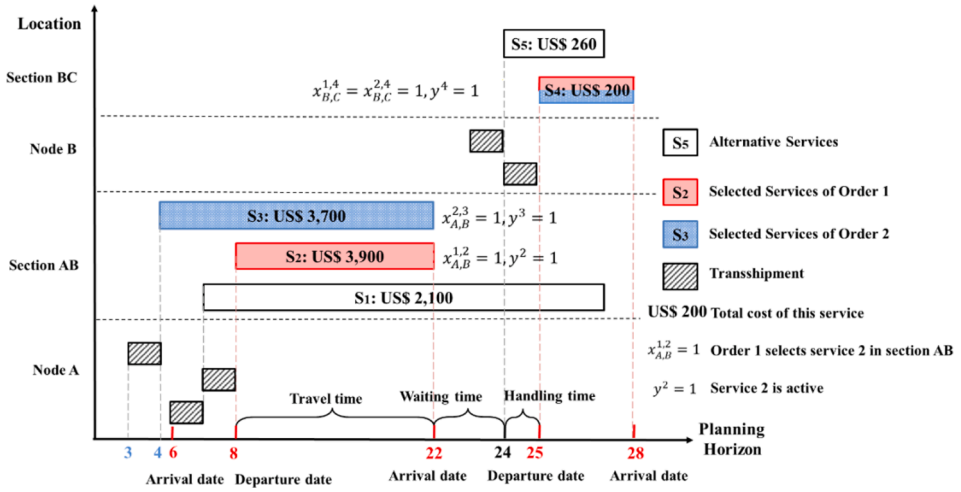


Fig. 15. An illustration of the logical relationship of two decision variables x_{ij}^{ks} and y^s

applied to the question of whether the transport schedule can be improved. The case study concerns service improvement of the CR Express (Wuhan-Hamburg) by adjusting trains' departure date.

6.1. Significance of the service departure date

As is shown in Fig. 6, regions like Wuhan (13) are predominantly maritime service users, while they become more likely to choose CR Express when transporting higher value cargoes or if the carbon price goes up. A question that may be raised is if there is any method to improve the market attractiveness of CR Express except cutting prices.

Most of the papers cited from the literature take an aggregate, macroscopic approach. A few examples, however, do consider a microscopic approach with respect to transport scheduling. Reinhardt et al. (2020) provide an example of a connection miss from the Panama rail service to the Caribbean maritime service, where cargoes have to wait several days if they miss the weekly maritime shipping. In this section, adjustments of a few days are evaluated to improve the attractiveness of the rail mode.

Fig. 14 **(a)** presents an example of a connection miss. To get a faster service plan like Fig. 14 **(b)**, the subsequent CR Express train should depart earlier to reduce the waiting time at transshipment node B, decreasing the in-transit inventory cost at the same time. Alternatively, this train could also depart later to provide enough handling time for railway collection service as shown in Fig. 14 **(c)**. This reduced carbon service plan may well offer lower transportation costs with lower carbon costs, as previously shown in Fig. 10 **(b)**. The second maritime shipping service (Shanghai-Koper) combining a slower inland waterway service makes a cheaper sequence than the first maritime shipping service (Shanghai-Koper) that departs earlier. A tiny relaxation of due time may help to save considerable transport cost.

6.2. Departure time optimization model

The new binary variable y^s is introduced based on the CMCFSSP to present whether service s is selected. The main idea of this enhancement model is finding optimal services from an alternative service set. In other words, the new variable means whether operators should provide services at h^s day.

Fig. 15 illustrates the relationship between the new variable y^s and the CMCFSSP decision variable x_{ij}^{ks} in previous sections but in a more microscopic perspective. There are two orders and five services (S_1, S_2, S_3, S_4, and S_5) available in section AB and its successor section BC. Each service is represented by an outlined rectangle (x-axis projection is its travel time, and y-axis projection is its occupied section), while the selected services are given a color fill.

As shown in Fig. 15, if service s is active, it is possible for order k to select it and flow from node i to node j, as depicted by constraint (13). The number of selected (active) services in the arc (i,j) should be less than the number of orders that flow via this section as described in constraint (14). The equation is special case, which only happens when each order selects an individual service.

$$x_{ij}^{ks} - y^s \leqslant 0, \forall k \in K, (i,j) \in A^k, s \in S_{ij} \tag{13}$$

$$\sum_{s \in S_{ij}} y^s - \sum_{k \in K | (i,j) \in A^k} \sum_{s \in S_{ij}} x_{ij}^{ks} \leqslant 0, \forall (i,j) \in A \tag{14}$$

These two constraints provide the lower bound and the upper bound of y^s, respectively, and make it possible for each order k to select individual or common service s in one physical arc (i,j).

A new capacity constraint (15) is given to present the restriction of the scheduled operation plan. To obtain the best service plan, a

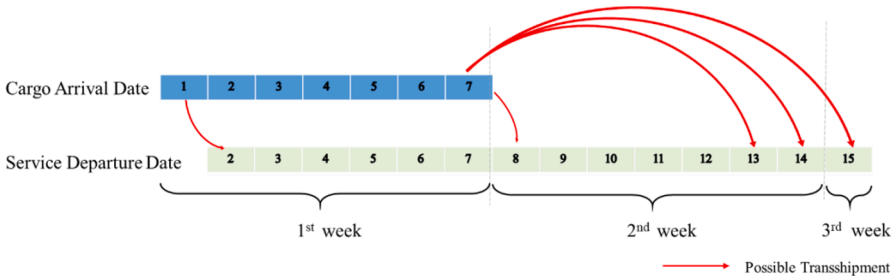


Fig. 16. Departure time of alternative services for a week demand

Table 8
Operation schedules for CR Express (Wuhan-Hamburg) in this section.

Schedules	Cost (US$/FEU)	Transit time (day)	Distance (km)	Capacity (FEU)	Departure date (n^{th} day)
Two departures a week	3950	15	11,880	45	1, 4
Four departures a week	3950	15	11,880	45	1, 3, 5, 7
Two departures a day	3950	15	11,880	45	1, 2, 3, 4, 5, 6, 7

wide range of alternative services are input, while the number of selected services shouldn't exceed the operators supplied. $w_{i,j}$ is the maximum number of services in the arc (i,j) in each period. For example, as for the test with the two-departure-a-week frequency, the number of alternative trains for a week should be 14 (two alternative trains every day) to cover the extreme situation where two trains depart on the same day. $w_{i,j}$ is equal to 2, which means at most 2 trains could be selected in 14 alternatives.

$$\sum_{s\in S_{ij}} y^s \leqslant w_{ij}, \forall (i,j) \in A \tag{15}$$

As cargoes arrived on Monday need another day to finish transshipment, their earliest possible departure days are Tuesday. To avoid losing orders, alternative services for a week demand are supposed to span from the first Tuesday to the Monday two weeks later, as shown in Fig. 16.

Considering that the current operation plan of CR Express is a weekly schedule, and the alternative services span in 3 weeks, services in this test are divided into 14 groups, where $S^{mn} \subseteqq S$. The numerical index presents the service set of the nth day in the mth week. If $m \in \{2,3\}$, then $n = 1$. If $m \in \{1,2\}$, then $n \in \{2,3,4,5,6,7\}$.

$$\sum_{s\in S^{1n}} y^s = \sum_{s\in S^{2n}} y^s, \forall n \in \{2,\ldots,7\} \tag{16}$$

$$\sum_{s\in S^{21}} y^s = \sum_{s\in S^{31}} y^s \tag{17}$$

Constraints (16) and (17) enforce periodicity, that the departure date for the same kind of services in different weeks are the same. For instance, CR Express train (Wuhan-Hamburg) departs on Tuesday in the first week, and it has to depart on Tuesday in the second week. Constraint (18) defines the range of the decision variable y^s.

$$y^s \in \{0,1\}, \forall s \in S \tag{18}$$

6.3. Numerical test

Data for this case is drawn from another 100 orders with the same distribution as the previous random case. This numerical test is solved by CPLEX to find the global optimal service sequences for cargoes transported from Wuhan in the region (13) to Hamburg. There are two groups of tests: a low-frequency service, where trains depart every Monday and Thursday in the initial operation plan, and a high-frequency operation plan, where trains depart every Monday, Wednesday, Friday, and Sunday as shown in Table 8.

The service network consists of 16 nodes, 28 physical arcs, and 350 alternative services. There are 3,727 constraints and 1,025,655 variables for both low and high frequency tests. To cover all potential situations, the service pool consists of 28 trains, two departures each day for two scheduling weeks.

For all instances, the total cost decrease after adjustments as shown in Table 9. In the low-frequency test, the smaller number of orders selecting CR Express (Wuhan-Hamburg) with lower in-transit inventory cost depicts that trains' operation plan set departure time earlier and find target customers (orders) more accurately. As for the high-frequency test, the higher transportation and carbon cost with lower in-transit inventory cost means faster but more expensive services are selected. Regarding the number of operating trains, it should be noted that the higher frequency leads to the higher service capacity, which attracts more cargo. What's more, departure time adjustments take better advantage of logistics resources, improving capacity utilization. This is mainly a result of the

Table 9
Service selection and cost breakdown for both tests before or after optimization.

Result	Low-frequency Test		High-frequency Test	
	Initial	Optimal	Initial	Optimal
No. of orders selecting CRE-WH*	15	11	20	22
No. of orders selecting other CR Express	19	22	14	13
No. of orders selecting maritime	66	67	66	65
Departure Date of CRE-WH*	1,4	3,3	1,3,5,7	3,3,4,5
Capacity utilization (%)	99.00	99.50	95.50	96.75
Transportation Cost (US$)	8,082,817	8,070,150	7,964,062	8,004,442
In-transit inventory Cost (US$)	3,488,811	3,481,995	3,530,207	3,457,509
Carbon Cost (US$)	528,175	527,492	529,428	532,411
Total Cost (US$)	12,099,803	12,079,637	12,023,697	11,994,362

* CRE-WH is short for CR Express (Wuhan-Hamburg) here.

better match for the delivery time, especially for the high value cargoes.

7. Discussion and managerial insights

A number of papers study shippers' preference of transportation mode, while "cost" is always the most important factor when making freight choice decisions (Li et al., 2020). Transport costs in this paper are end-user prices, so it seems that transport cost covers a high proportion of the total cost, about 80%, as shown in Table 6. The regional differences of transport service price (caused by local competitiveness) result in an odd patchwork of geographic preferences for transport mode, instead of a continuous dividing line given by Lu et al. (2019). Economic development in the inland regions of northwestern China will clearly favor CR Express.

High value cargoes prefer shorter transit time, so the faster CR Express trains or the shorter voyages shipping remains the primary choice. Jiang et al. (2018) point out that IT products are much easier to select CR Express than other cargoes. On the contrary, Yin et al. (2020) ignore the effect of time value and got an improper result that no shipper would select CR Express on the east coast without government subsidies. Under the double influence of transit time and carbon emissions, reducing driving distance for the "last mile" transport is the primary choice of high value cargoes, particularly in the event of a high carbon price.

We believe that shippers, including cargo owners and freight forwarders, should pay attention to service departure time besides service cost and delivery time as shown in Fig. 14. On one hand, optimal service connections with suitable departure dates make it possible to reduce waiting time at intermediate ports or railway stations to speed delivery up (Reinhardt et al., 2020). On the other hand, a later departure provides enough collection time for a slower but cheaper and environmental-friendly service. Thus, a small relaxed delivery time limitation may lead to a much lower transportation cost. It would require communication between transport providers and shippers. The two parties may not realize how close they are to a solution that offers value with a minor change.

We also believe that both maritime shipping and CR Express operators should focus on service improvement and find the right target customers, instead of trying to grasp more cargo from one other by direct competition, such as price competition. The core elements of selecting logistics services are due time and cargo value, as described in Fig. 7 and Fig. 9. Deep-sea shipping has obvious advantages in planned, long-term supply and low carbon emission, while CR Express is more valuable for urgent demands like bulk emergency aid or higher value cargoes. Fig. 8 and Fig. 9 also show that CR Express has not yet fully utilized its geographical advantage, especially the Central Corridor that connects Europe via inner Mongolia (region 7). The middle regions of north China, such as region 8, may find CR Express more attractive after optimizing operating plans.

Our results could also be an instrument for tackling climate change. As stated earlier, a suitable carbon policy, such as a carbon price, will encourage a shift to lower carbon emission service plans (Jiang et al., 2018). Surprisingly, the "last mile" transport links have much more influence over the transport mode and carbon emissions than the long-haul transport mode, as shown in Fig. 12**and** Fig. 13. Our results may be useful for the Chinese government and railway companies. The geographical preference line of CR Express would be closer to the coast by investing in appropriate fundamental facilities and optimizing operation timetables.

8. Conclusions

This paper developed a decision support tool for shippers to select appropriate transport services in the context of BRI. The CMCFSSP model has been developed and applied to numerical studies with recent, realistic scheduled transport services in the Silk Road corridors. Multiple commodity flows are depicted by a capacitated multigraph, where arc costs include transport cost, in-transit inventory cost and carbon impact. We believe that this model and the associated application improve upon the state of the art on models and methods for the BRI.

Computational results and sensitivity analyses prove some statements stated in previous work, but also produce a set of counter-intuitive findings. As expected, transportation costs usually account for a larger percentage of the total cost, exceeding inventory and carbon emission costs. High value cargoes or orders with tight delivery times prefer CR Express trains. Maritime logistics services remain the most cost-effective options for the older developed regions of China around port cities. However, developed regions further away from the coast or with weaker "last mile" connections to ports find the rail mode more competitive. In spite of the very low carbon emission rates of modern maritime shipping, it would seem that the faster inland services (CR Express) are sometimes superior

Table A
Alternative services for six orders in the small-sized test.

Set	No.	From node	To node	Departure date	Cost (US $/FEU)	Transit time (Day)	Distance (km)	Capacity (FEU)	Carbon coefficient (×10-3 ton/(FEU·km)
Collection Services									
Order 1	1	Beijing	Tianjin	1	206	2	121	45	0.902
	2	Beijing	Tianjin	1	342	1	137	100	1.51
	3	Beijing	Jinan	1	381	3	406	45	0.902
	4	Beijing	Jinan	1	530	2	422	100	1.51
	5	Beijing	Xi'an	2	807	4	1283	45	0.902
	6	Beijing	Xi'an	1	1067	3	1081	100	1.51
	7	Beijing	Shenyang	1	632	3	589	45	0.902
	8	Beijing	Shenyang	1	684	2	642	100	1.51
Order 2	9	Yinchuan	Qingdao	1	968	2	1456	100	1.51
	10	Yinchuan	Qingdao	1	1221	3	1625	45	0.902
	11	Yinchuan	Tianjin	1	1000	2	1166	100	1.51
	12	Yinchuan	Tianjin	2	1035	3	1472	45	0.902
	13	Yinchuan	Shanghai	1	1704	2	1951	100	1.51
	14	Yinchuan	Shanghai	2	1621	4	2215	45	0.902
	15	Yinchuan	Lanzhou	1	386	1	468	45	0.902
	16	Yinchuan	Lanzhou	1	569	1	426	100	1.51
	17	Yinchuan	Xi'an	1	588	2	692	100	1.51
	18	Yinchuan	Xi'an	1	619	3	806	45	0.902
	19	Yinchuan	Urumchi	1	1043	1	1191	100	1.51
	20	Yinchuan	Urumchi	1	1421	3	2060	45	0.902
Order 3	21	Shanghai	Suzhou	1	100	1	104	1000	1.51
	22	Shanghai	Suzhou	2	138	1	84	1000	0.902
	23	Shanghai	Yiwu	1	140	1	284	1000	1.51
	24	Shanghai	Hefei	1	553	1	449	1000	0.902
	25	Shanghai	Hefei	1	453	1	567	1000	1.51
Order 4	26	Zhengzhou	Lianyungang	1	449	1	349	1000	1.51
	27	Zhengzhou	Shanghai	1	836	1	942	1000	1.51
	28	Zhengzhou	Shanghai	1	744	2	998	1000	0.902
	29	Zhengzhou	Ningbo	1	894	1	1083	1000	1.51
	30	Zhengzhou	Ningbo	1	893	1	1300	1000	0.902
	31	Zhengzhou	Urumchi	1	1543	5	3140	1000	0.902
Order 5	32	Wuhan	Shenzhen	1	1065	3	1171	1000	1.51
	33	Wuhan	Shanghai	1	310	4	820	1000	0.902
	34	Wuhan	Shanghai	1	90	7	704	1000	0.275
	35	Wuhan	Shanghai	1	758	1	767	1000	1.51
	36	Wuhan	Ningbo	1	937	2	874	1000	1.51
	37	Wuhan	Ningbo	1	100	8	956	1000	0.275
	38	Wuhan	Ningbo	1	781	4	923	1000	0.902
	39	Wuhan	Urumchi	1	2371	4	3267	1000	0.902
Order 6	40	Guiyang	Shenzhen	1	1084	3	1060	1000	1.51
	41	Guiyang	Shenzhen	1	1504	4	959	1000	0.902
	42	Guiyang	Guangzhou	1	542	3	930	1000	0.902
	43	Guiyang	Guangzhou	1	1089	1	867	1000	1.51
	44	Guiyang	Qinzhou	1	820	1	728	1000	1.51
	45	Guiyang	Qinzhou	1	732	2	768	1000	0.902
	46	Guiyang	Urumchi	1	3969	3	3352	1000	1.51
	47	Guiyang	Urumchi	2	2622	5	3578	1000	0.902
	48	Guiyang	Chengdu	1	677	3	646	1000	0.902
	49	Guiyang	Chengdu	1	568	2	646	1000	1.51
Intercontinental long-haul transport service									
Maritime services	50	Tianjin	Piraeus	3	2100	38	16,175	10,000	0.275
	51	Tianjin	Piraeus	10	2100	38	16,175	10,000	0.275
	52	Tianjin	Hamburg	3	2492	41	23,596	10,000	0.275
	53	Tianjin	Hamburg	10	2492	41	23,596	10,000	0.275
	54	Qingdao	Hamburg	5	2690	37	22,998	10,000	0.275
	55	Qingdao	Hamburg	12	2690	37	22,998	10,000	0.275
	56	Shanghai	Hamburg	7	2524	35	22,737	10,000	0.275
	57	Shanghai	Hamburg	14	2524	35	22,737	10,000	0.275
	58	Shanghai	Copenhagen	1	2800	43	21,011	1,000	0.275
	59	Lianyungang	Piraeus	3	2400	28	14,910	1,000	0.275
	60	Shanghai	Piraeus	4	2250	26	15,072	1,000	0.275
	61	Ningbo	Piraeus	3	2300	26	14,938	1,000	0.275
	62	Shenzhen	Koper	5	2300	27	15,075	1,000	0.275

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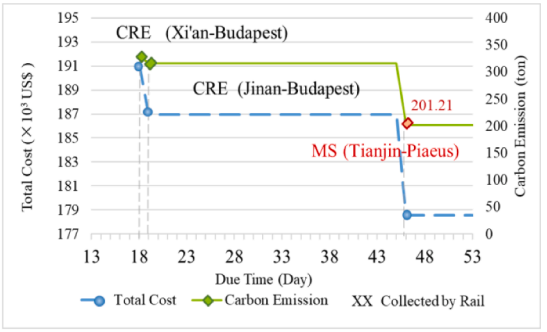
Table A (*continued*)

Set	No.	From node	To node	Departure date	Cost (US $/FEU)	Transit time (Day)	Distance (km)	Capacity (FEU)	Carbon coefficient (×10-3 ton/(FEU·km)
	63	Shanghai	Koper	6	2750	36	16,364	1,000	0.275
	64	Shanghai	Koper	13	2750	36	16,364	1,000	0.275
	65	Ningbo	Koper	4	2900	33	16,231	1,000	0.275
	66	Ningbo	Koper	11	2900	33	16,231	1,000	0.275
	67	Shenzhen	Algeciras	6	2350	28	16,236	1,000	0.275
	68	Guangzhou	Algeciras	5	2530	35	8,472	1,000	0.902
	69	Qinzhou	Algeciras	4	2300	27	16,040	1,000	0.275
CR Express	70	Jinan	Budapest	4	4200	15	9,580	45	0.902
	71	Jinan	Budapest	10	4200	15	9,580	45	0.902
	72	Xi'an	Budapest	5	3800	13	9,000	45	0.902
	73	Xi'an	Budapest	12	3800	13	9,000	45	0.902
	74	Shenyang	Hamburg	3	3900	13	9,985	45	0.902
	75	Shenyang	Hamburg	6	3900	13	9,985	45	0.902
	76	Lanzhou	Hamburg	3	4150	15	8,027	45	0.902
	77	Lanzhou	Hamburg	6	4150	15	8,027	45	0.902
	78	Xi'an	Hamburg	3	3850	13	9,440	45	0.902
	79	Xi'an	Hamburg	6	3850	13	9,440	45	0.902
	80	Urumchi	Hamburg	3	4200	10	6,812	45	0.902
	81	Urumchi	Hamburg	6	4200	10	6,812	45	0.902
	82	Urumchi	Hamburg	9	4200	10	6,812	45	0.902
	83	Suzhou	Duisburg	4	4100	15	11,220	45	0.902
	84	Yiwu	Minsk	3	4800	12	10,370	1000	0.902
	85	Hefei	Hamburg	3	4100	15	10,647	45	0.902
	86	Zhengzhou	Munich	1	4253	14	10,214	1000	0.902
	87	Urumchi	Lodz	8	3810	10	6007	1000	0.902
	88	Wuhan	Pardubice	3	4400	14	10,100	1000	0.902
	89	Urumchi	Nuremberg	6	3800	10	8000	1000	0.902
	90	Wuhan	Lodz	1	4200	15	10,800	1000	0.902
	91	Urumchi	Madrid	5	2500	19	8842	1000	0.902
	92	Urumchi	Madrid	8	2500	19	8842	1000	0.902
	93	Chengdu	Tilburg	5	3800	14	18,455	1000	0.902
Distribution services									
Order 1	94	Piraeus	Budapest	42	1400	4	1400	1000	0.902
	95	Piraeus	Budapest	42	4780	3	1474	1000	1.51
	96	Piraeus	Budapest	50	1400	4	1400	1000	0.902
	97	Piraeus	Budapest	49	4780	3	1474	1000	1.51
	98	Hamburg	Budapest	46	1200	4	1200	1000	0.902
	99	Hamburg	Budapest	53	1200	3	1163	1000	0.902
	100	Hamburg	Budapest	17	1200	4	1200	1000	0.902
	101	Hamburg	Budapest	20	1200	4	1163	1000	0.902
	102	Hamburg	Budapest	46	3349	3	1200	1000	1.51
	103	Hamburg	Budapest	52	3349	3	1163	1000	1.51
	104	Hamburg	Budapest	17	3349	3	1200	1000	1.51
	105	Hamburg	Budapest	20	3349	3	1163	1000	1.51
Order 2	–	–	–	–	–	–	–	–	–
Order 3	107	Hamburg	Copenhagen	45	1527	1	332	1000	1.51
	108	Hamburg	Copenhagen	45	320	2	320	1000	0.902
	109	Duisburg	Copenhagen	22	680	2	680	1000	0.902
	110	Duisburg	Copenhagen	22	3220	1	700	1000	1.51
	111	Minsk	Copenhagen	16	4429	1	963	1000	0.902
	112	Minsk	Copenhagen	17	1000	2	1000	1000	1.51
Order 4	113	Piraeus	Athens	32	45	1	9	1000	1.51
	114	Piraeus	Athens	32	10	1	10	1000	0.902
	115	Munich	Athens	16	2050	4	2,050	1,000	0.902
	116	Munich	Athens	16	4,361	3	2,035	1,000	1.51
	117	Lodz	Athens	19	2,248	5	2,248	1,000	0.902
	118	Lodz	Athens	19	10,342	3	2,248	1,000	1.51
Order 5	119	Koper	Prague	33	814	2	814	1,000	0.902
	120	Koper	Prague	43	814	2	814	1,000	0.902
	121	Koper	Prague	50	814	2	814	1,000	0.902
	122	Koper	Prague	38	814	2	814	1,000	0.902
	123	Koper	Prague	45	814	2	814	1,000	0.902
	124	Pardubice	Prague	18	542	1	118	1,000	1.51
	125	Pardubice	Prague	19	120	1	118	1,000	0.902
	126	Nuremberg	Prague	19	300	2	295	1,000	0.902
	127	Lodz	Prague	17	542	2	542	1,000	0.902
Order 6	128	Algeciras	Madrid	35	2,400	2	648	1,000	1.51

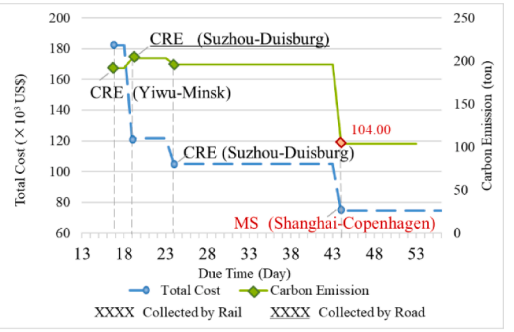
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Table A (*continued*)

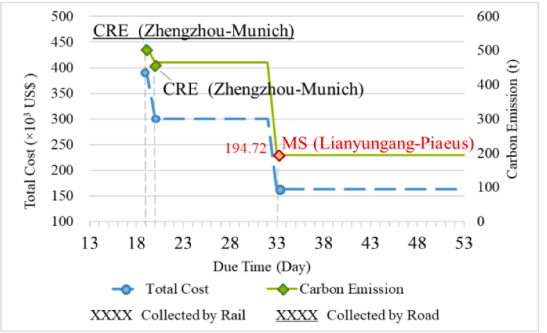
Set	No.	From node	To node	Departure date	Cost (US $/FEU)	Transit time (Day)	Distance (km)	Capacity (FEU)	Carbon coefficient (×10-3 ton/(FEU·km)
	129	Algeciras	Madrid	36	60	3	650	1000	0.902
	130	Algeciras	Madrid	41	600	3	650	1000	0.902
	131	Algeciras	Madrid	32	600	3	650	1000	0.902
	132	Tilburg	Madrid	20	7741	3	1683	1000	1.51
	133	Tilburg	Madrid	21	1680	3	1680	1000	0.902



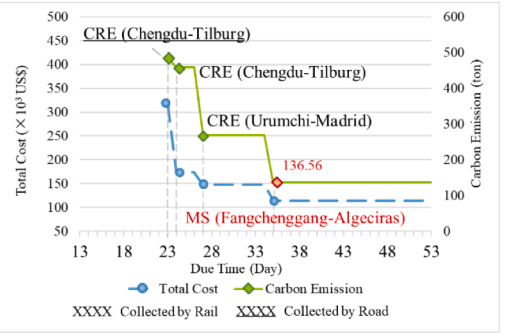
(a) Total cost and carbon emission for order 1



(b) Total cost and carbon emission for order 3



(c) Total cost and carbon emission for order 4



(d) Total cost and carbon emission for order 6

Fig. B. Total cost and carbon emission of optimal solutions in the small-sized test with a range of due time

in reducing carbon emissions. The greatest factor in carbon emissions is the "last mile" transport connection. Access to collection terminals for either rail or maritime services is a deciding factor in mode selection and carbon emissions. Flexibility and frequency of departures is the next factor in the choice of mode. A few days' mismatch between the arrival date of a shipment and the service departure date can result in the selection of the competing service. Results in this paper show much more complexity and variety in the modal selection than previous papers, providing an odd patchwork of geographic preferences for CR Express.

Last but not least, we believe that the tools presented in this paper can also be useful from a policy perspective, particularly as regards the drive to obtain greener solutions for international intermodal transport services. In that sense, the models developed in this paper allow policymakers to explore appropriate carbon prices and to examine whether policies are moving towards greener solutions.

As the transport cost of CR Express is end-user prices in practice, this paper doesn't consider different holding cost at different locations or for different services. As it is, the data available was limited. Actual historical records of shippers at different locations can't be currently obtained. If more detailed data were available, this study could also be extended to a stochastic programming model that consider holding costs in different scenarios. More detailed data would allow distribution fitting and simulation of shipper behavior. It will also be interesting to test the CMCFSSP as a hierarchical multi-objective problem to discuss the impact of shippers' individual sensitivity to transport, inventory and carbon emission costs on their selection behavior and the geographic preferences of CR Express.

CRediT authorship contribution statement

Yingxiu Qi: Conceptualization, Investigation, Methodology, Software, Writing – original draft. **Steven Harrod:** Conceptualization, Methodology, Writing – review & editing. **Harilaos N. Psaraftis:** Formal analysis, Writing – review & editing. **Maoxiang Lang:** Project

Table C
The number of times for each order selecting CR Express in 100 replications.

Orders	Baseline case	Due date			In-transit inventory cost			Carbon cost		
	Min total cost	Min total cost			Min total cost			Min total cost		Min carbon cost
	$l^k \sim$ U(30, 60), $\beta^k \sim$U (10, 50), e = 27.35	l^k = 30, $\beta^k \sim$U(10, 50), e = 27.35	l^k = 40,$\beta^k \sim$ U(10, 50), e = 27.35	*l^k = 50, $\beta^k \sim$U(10, 50), e = 27.35	$l^k \sim$ U(30, 60), β^k=0, e = 27.35	$l^k \sim$ U(30, 60), β^k=27.12, e = 27.35	$l^k \sim$ U(30, 60),β^k = 45.21, e = 27.35	$l^k \sim$ U(30, 60), $\beta^k \sim$U (10, 50), e = 0	$l^k \sim$ U(30, 60), $\beta^k \sim$U (10, 50), e = 100	$l^k \sim$ U(30, 60), $\beta^k \sim$U (10, 50), e = 27.35
1	48	100	100	0	48	48	100	48	48	100
2	85	100	100	53	53	83	100	53	88	100
3	49	100	100	0	49	49	100	49	49	49
4	40	100	68	0	40	40	40	40	40	40
5	53	100	100	0	53	53	53	53	53	53
6	57	100	100	0	57	57	57	57	57	57
7	50	100	100	0	50	50	50	50	50	50
8	41	100	100	0	41	41	49	41	41	41
9	31	100	0	0	31	31	31	31	31	31
10	27	100	0	0	27	27	100	27	27	27
11	35	100	0	0	35	35	35	35	35	35
12	30	100	0	0	30	30	30	30	30	30
13	27	100	0	0	27	27	100	27	27	27
14	48	100	100	0	43	48	100	48	48	43
15	24	100	0	0	24	24	24	24	24	24
16	19	100	0	0	19	19	19	19	19	19
17	13	100	0	0	13	13	13	13	13	13
18	16	100	0	0	16	16	78	16	16	16
19	23	100	0	0	19	23	23	23	19	19
20	25	100	0	0	25	25	25	25	25	25
21	14	100	0	0	14	14	14	14	14	14
22	8	100	0	0	8	8	8	8	8	8
23	31	100	25	0	31	31	31	31	31	31
24	16	100	0	0	16	16	16	16	16	16
25	6	100	0	0	6	6	6	6	6	6
26	13	100	0	0	13	13	13	13	13	13
27	8	100	0	0	8	8	8	8	8	8
28	100	100	100	100	1	100	100	100	100	51
29	65	100	100	17	55	64	100	65	65	55
30	100	100	100	100	26	100	100	100	100	100
31	100	100	100	100	41	100	100	100	100	100
32	100	100	100	100	100	100	100	100	100	100
33	79	100	100	51	64	75	100	100	64	40
34	73	100	100	16	69	73	100	69	73	100

* The results ofl^k = 50, $\beta^k \sim$U(10, 50), e = 27.35 are the same withl^k = 60, $\beta^k \sim$U(10, 50), e = 27.35.

administration, Investigation, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. . Full table for alternative services in small data set

Table A shows the service pool of the small-sized test in Section 4.1, including individual collection and distribution services for each order and public services available for all customers. It can be seen that services are identified by origin, destination, departure date, cost, transit time, travel distance, service capacity and carbon emission. Note that, transport mode can be recognized by carbon coefficient that 1.51×10^{-3} ton/(FEU·km) is for road service, 0.902×10^{-3} ton/(FEU·km) for rail and 0.275×10^{-3} ton/(FEU·km) for water.

Appendix B. . The remaining sensitivity analysis of the small case

Fig. B shows the same test result with Fig. 10. Total cost and carbon emission of optimal solutions in the small-sized test with a range of due time carbon emission is fluctuant when we select different services.

Appendix C. . The percentage of selecting CR Express of all large-sized tests

Table C shows the preference of each order for CR Express in all large-sized tests. The first column is the order number. The second column is the basic case of large-sized test in Section 4.2, where the due date (l^k) are sampled uniformly from 30 to 60 days, the in-transit inventory cost (β^k) is drawn from a uniform distribution from US$ 10 to US$ 50/FEU per day, and the carbon rate is US$ 27.35/ton. The mode selection of each order is presented in Fig. 6.

The 3rd to 5th columns show the sensitivity analysis of due date. The only parameter changed is due date of each order, which is 30, 40, 50 and 60 days respectively. (The results of$l^k = 50$ are the same with$l^k = 60$, so Table C doesn't show the results of$l^k = 60$) Tight time limitation encourages orders to select CR Express, while relaxed due date makes it possible to choose slower but much cheaper maritime services. Considering slower delivery leads to higher in-transit inventory cost and long-distance transport increases carbon cost, some orders still prefer CR Express.

The sensitivity of in-transit inventory cost is in the 6th to 8th columns. The comparison between the 6th (β^k=0) and the 7th column (β^k=27.12) is given in Fig. 8, while the comparison between the 6th and the 8th column (β^k=45.21) is shown in Fig. 9.

The 9th to 11th columns on the right show the sensitivity analysis of carbon emission costs. The differences between the 9th column ($e = 0$) and baseline case ($e = 27.35$) are presented in Fig. 12, while Fig. 13 shows the difference between the 9th and the 10th column ($e = 100$). The results of the last column are presented in Fig. 11, whose objective function changed to minimize carbon emission cost only.

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